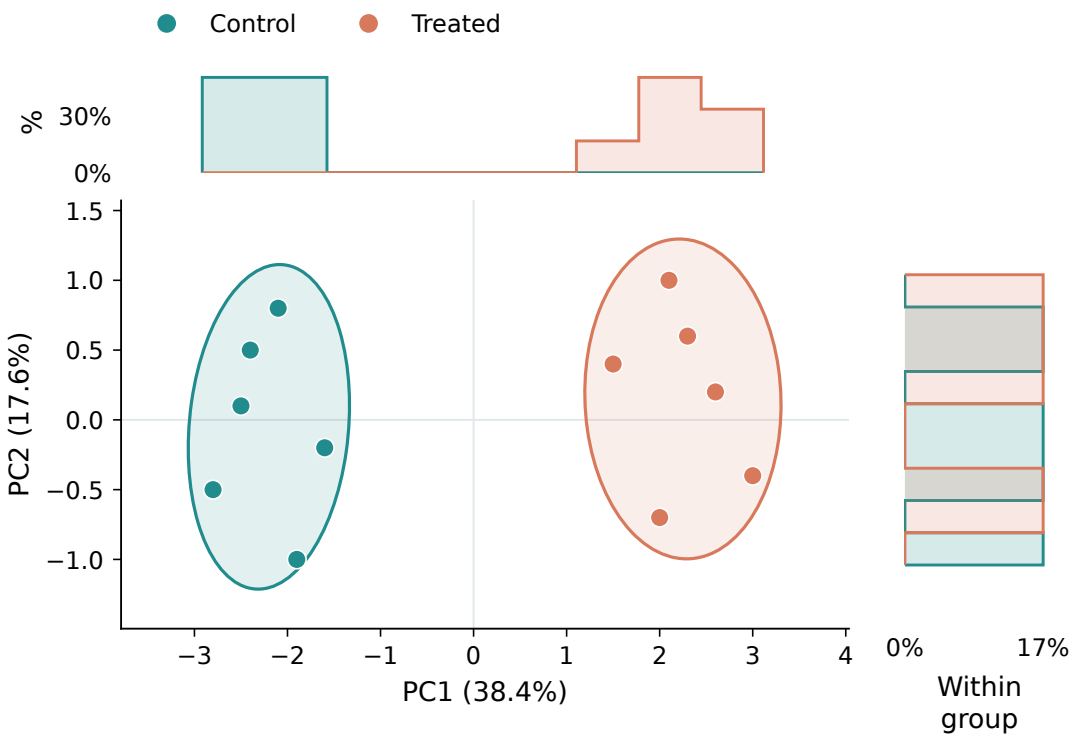
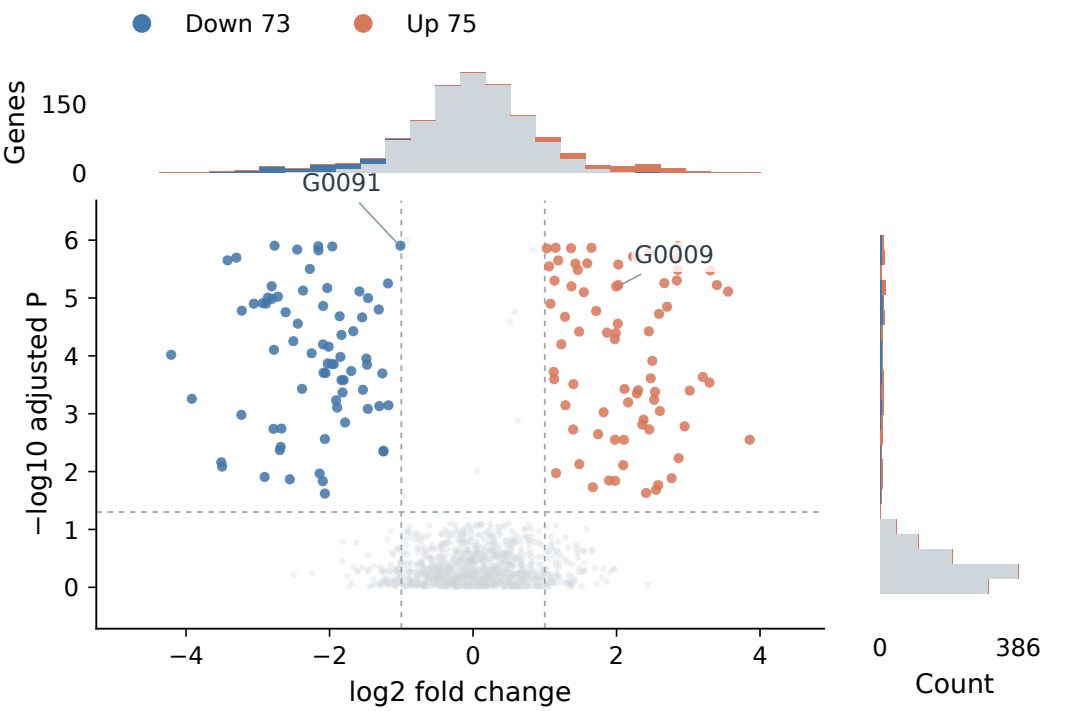
Treated versus Control / illustrative
12 synthetic donors / all results are fictional

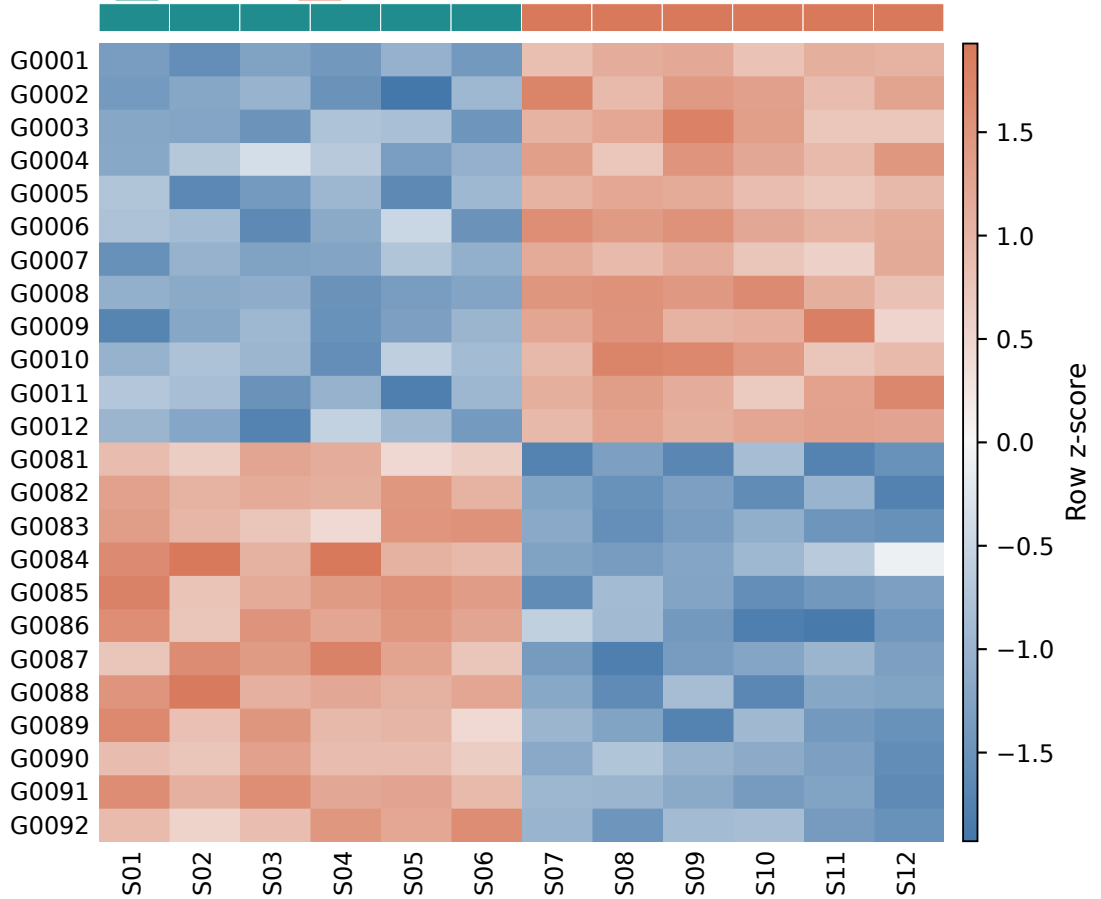
A Sample architecture



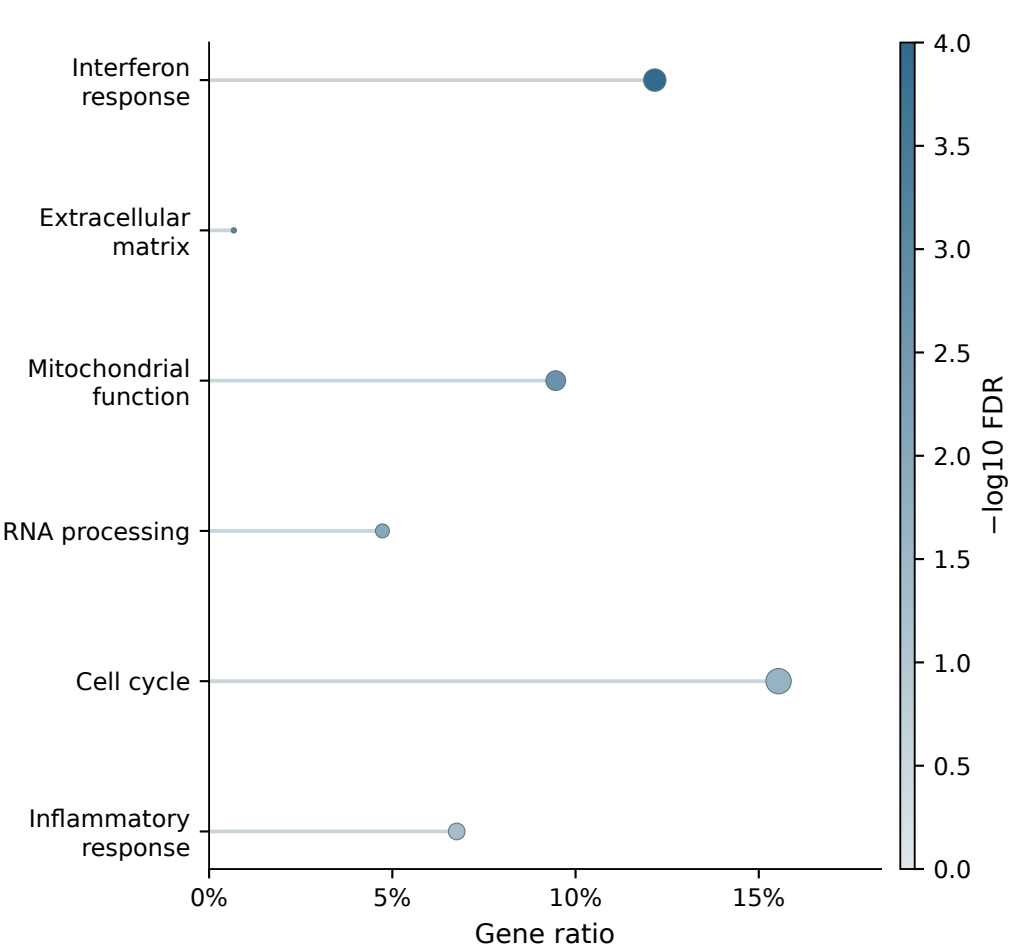
B Differential response

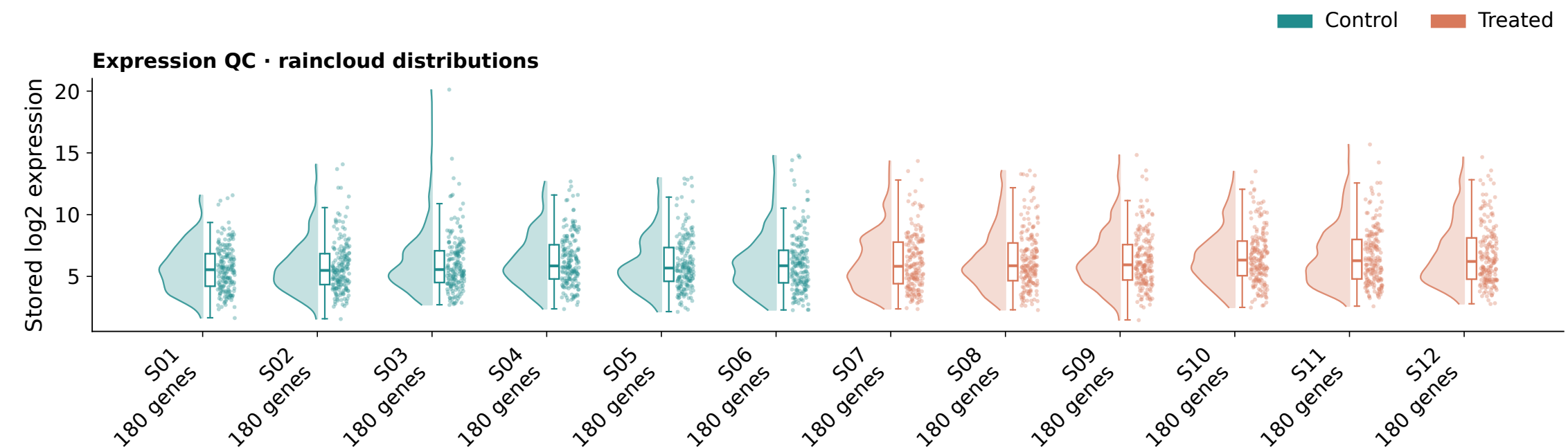


C Prespecified expression program

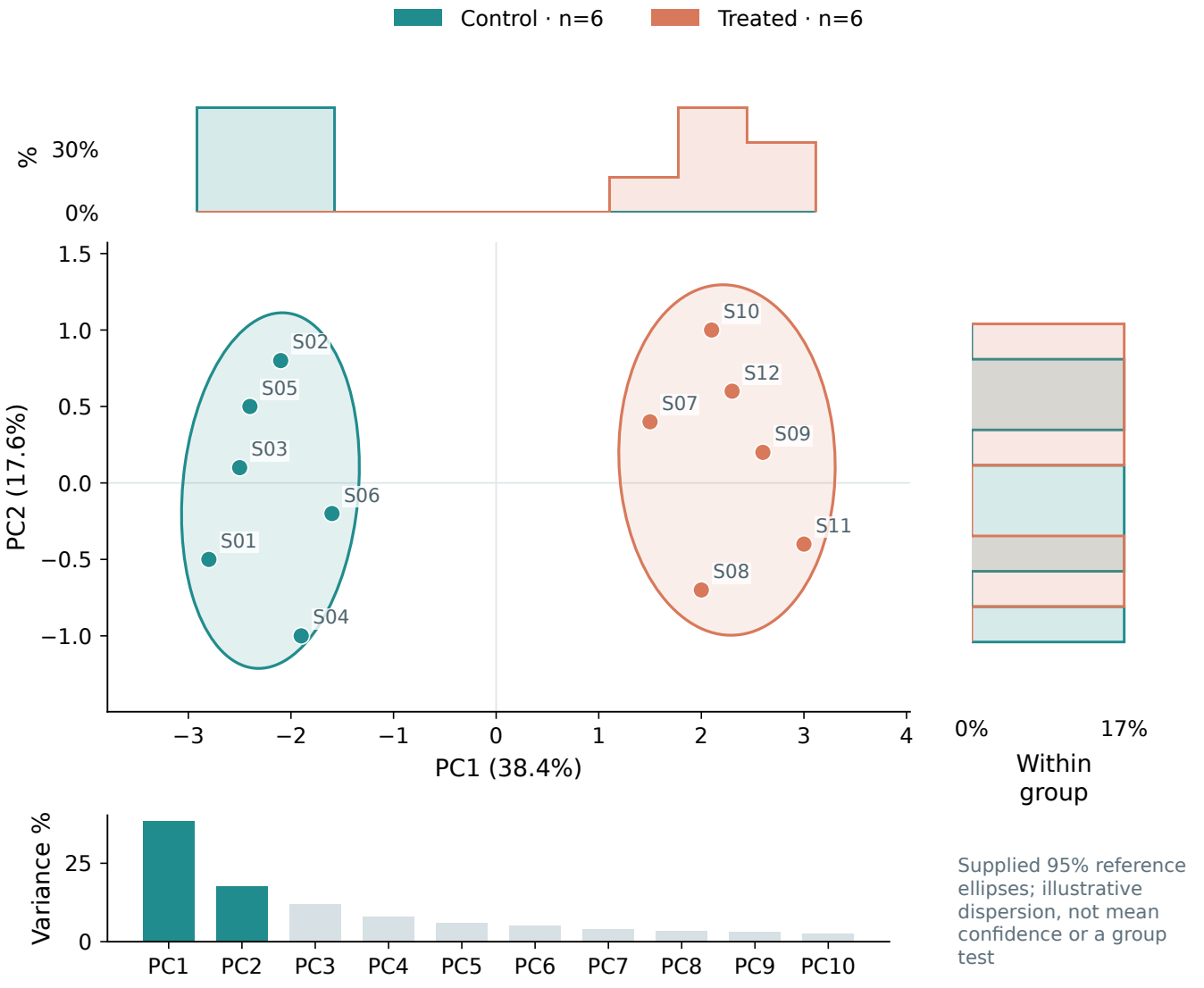


D Functional context

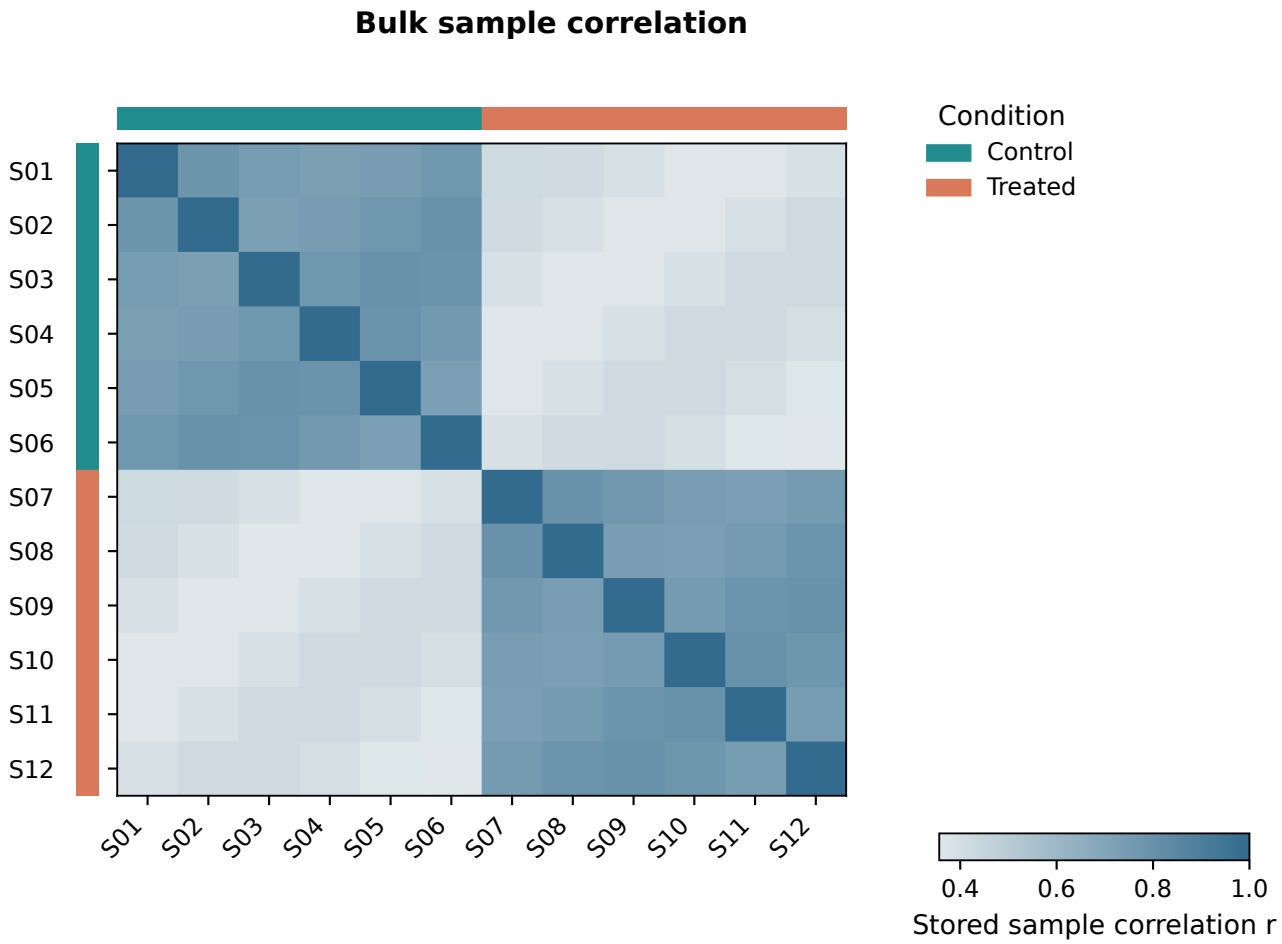


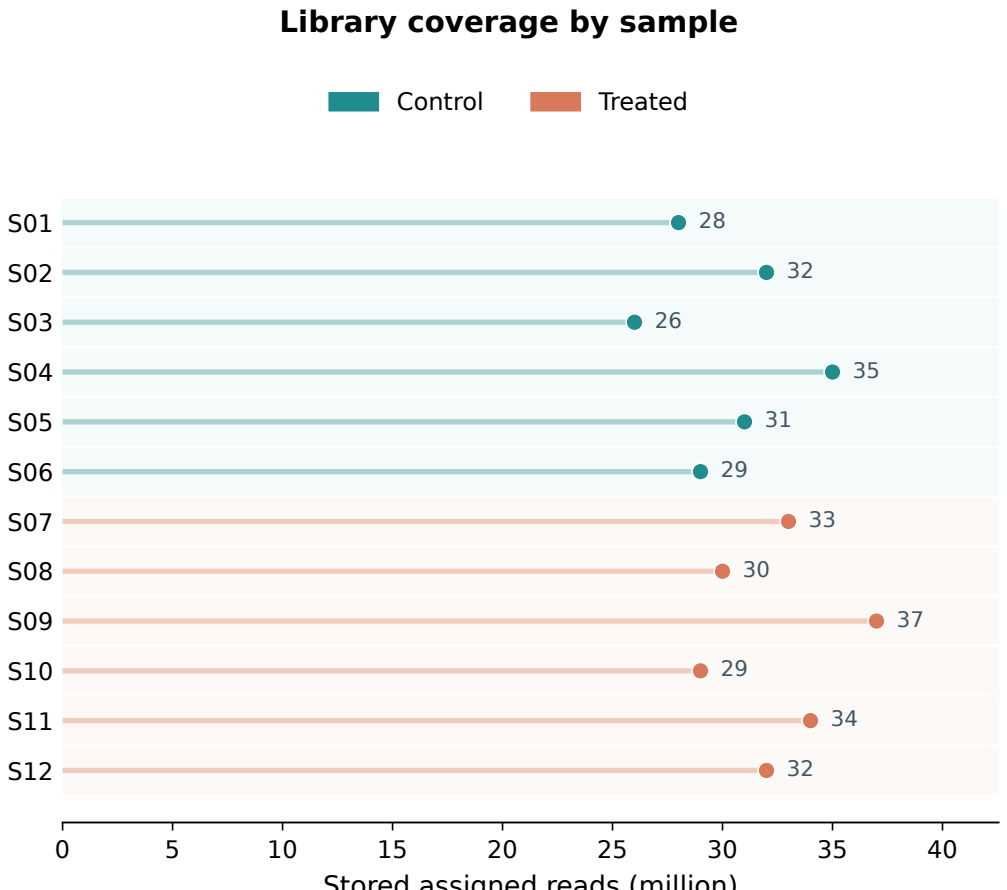

Control
Treated
Expression QC · raincloud distributions
Stored log2 expression
20
15
10
5
S01
180 genes
S02
180 genes
S03
180 genes
S04
180 genes
S05
180 genes
S06
180 genes
S07
180 genes
S08
180 genes
S09
180 genes
S10
180 genes
S11
180 genes
S12
180 genes

Sample structure · dispersion and marginal proportions



Supplied 95% reference ellipses; illustrative dispersion, not mean confidence or a group test

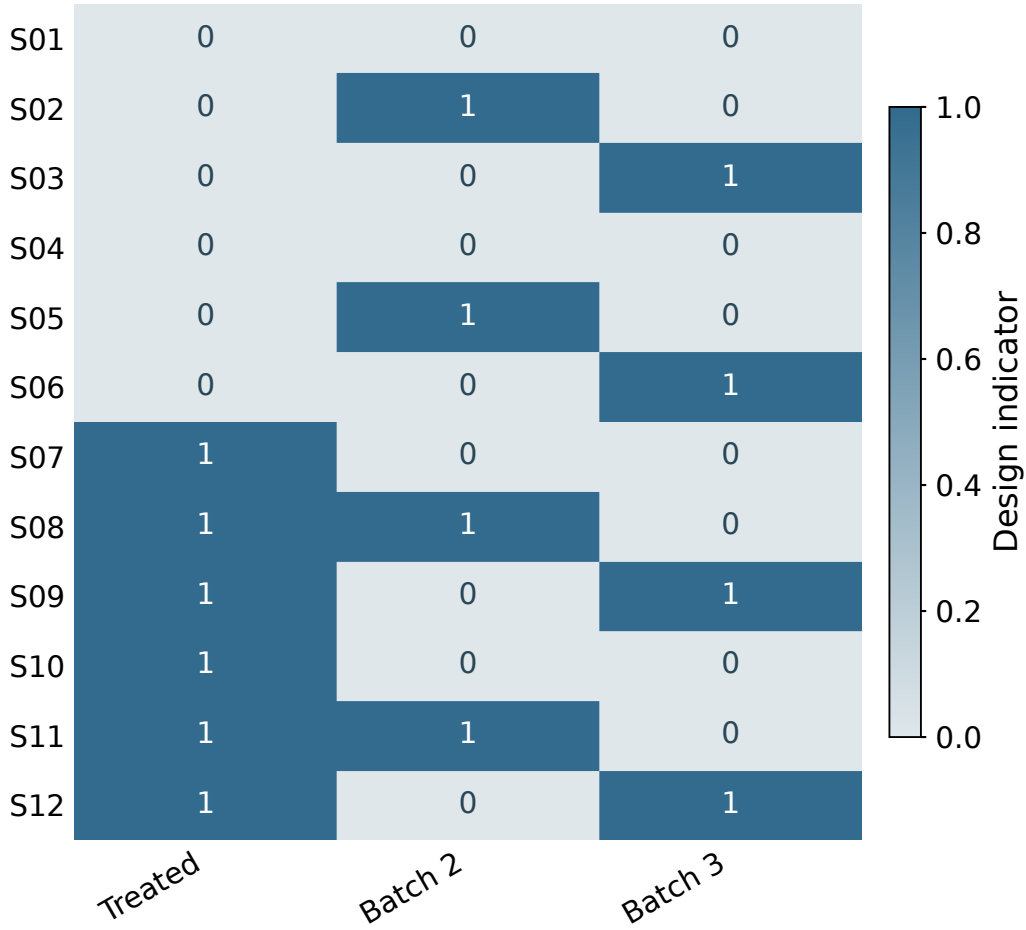

Bulk sample correlation
Condition
Control
Treated
S01
S02
S03
S04
S05
S06
S07
S08
S09
S10
S11
S12
0.4
0.6
0.8
1.0
Stored sample correlation r

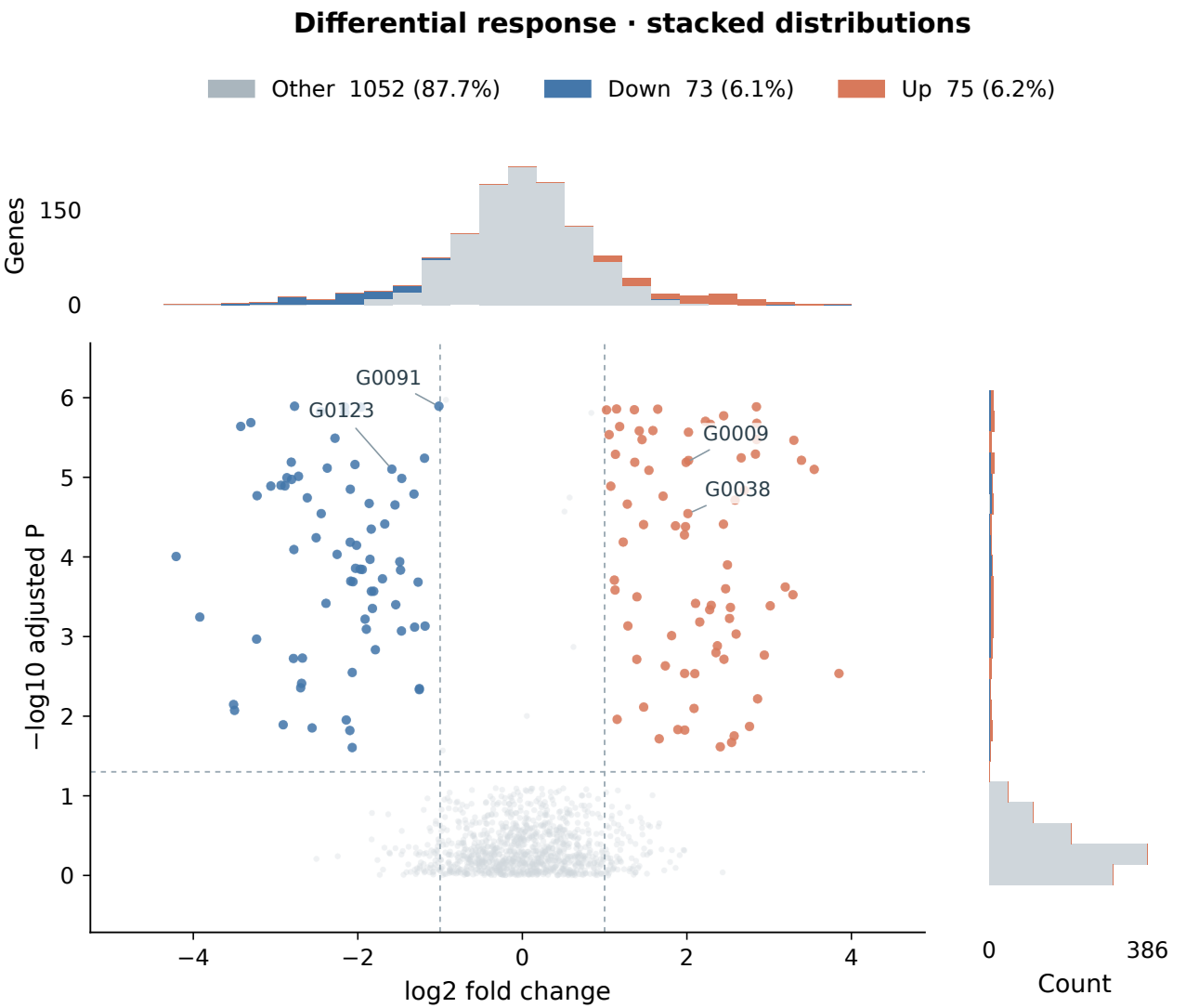

Library coverage by sample
Control
Treated
S01
S02
S03
S04
S05
S06
S07
S08
S09
S10
S11
S12
28
32
26
35
31
29
33
30
37
29
34
32
0
5
10
15
20
25
30
35
40
Stored assigned reads (million)

Donors, condition and batch
12 fictional donors · six per condition · supplied results only

A

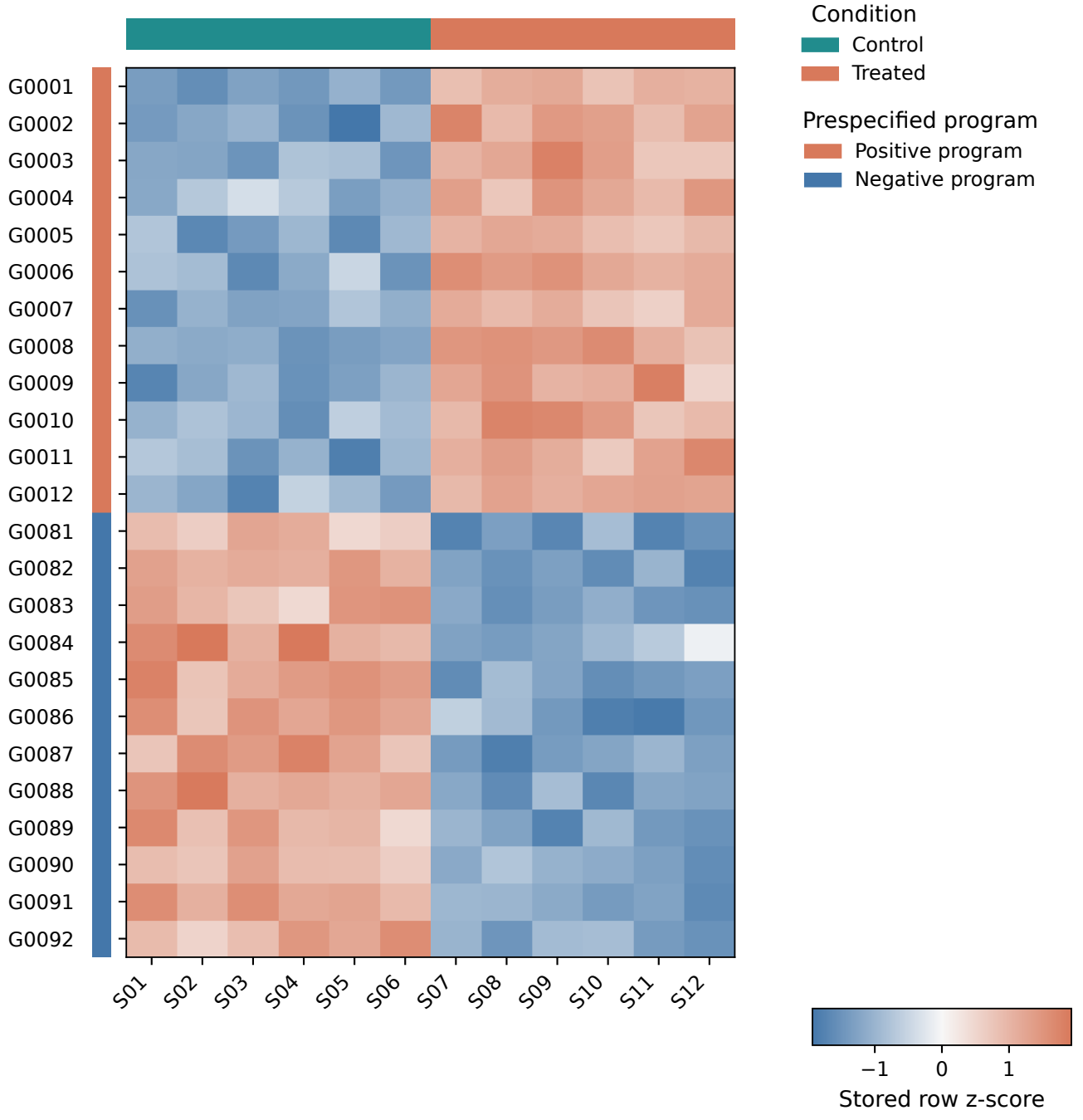
Declared design columns





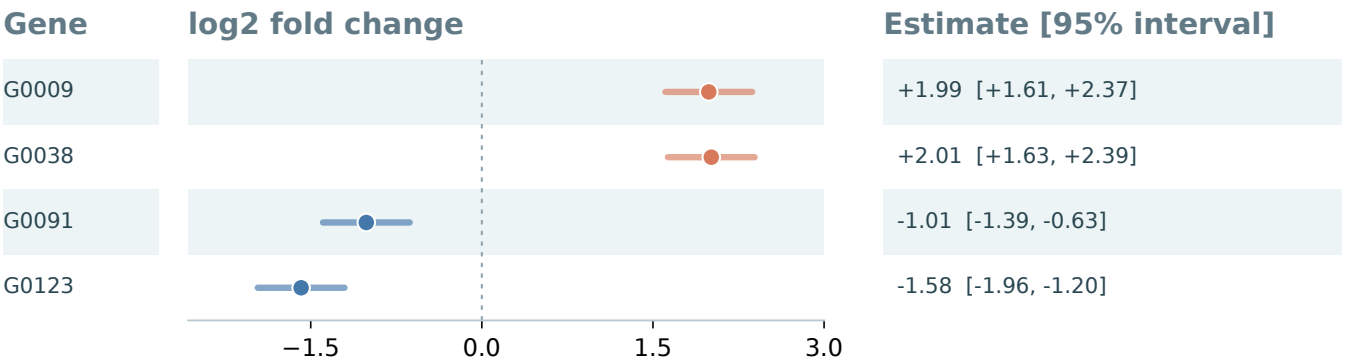
Stacked marginals retain all genes · FDR ≤ 0.05 and |log2FC| ≥ 1

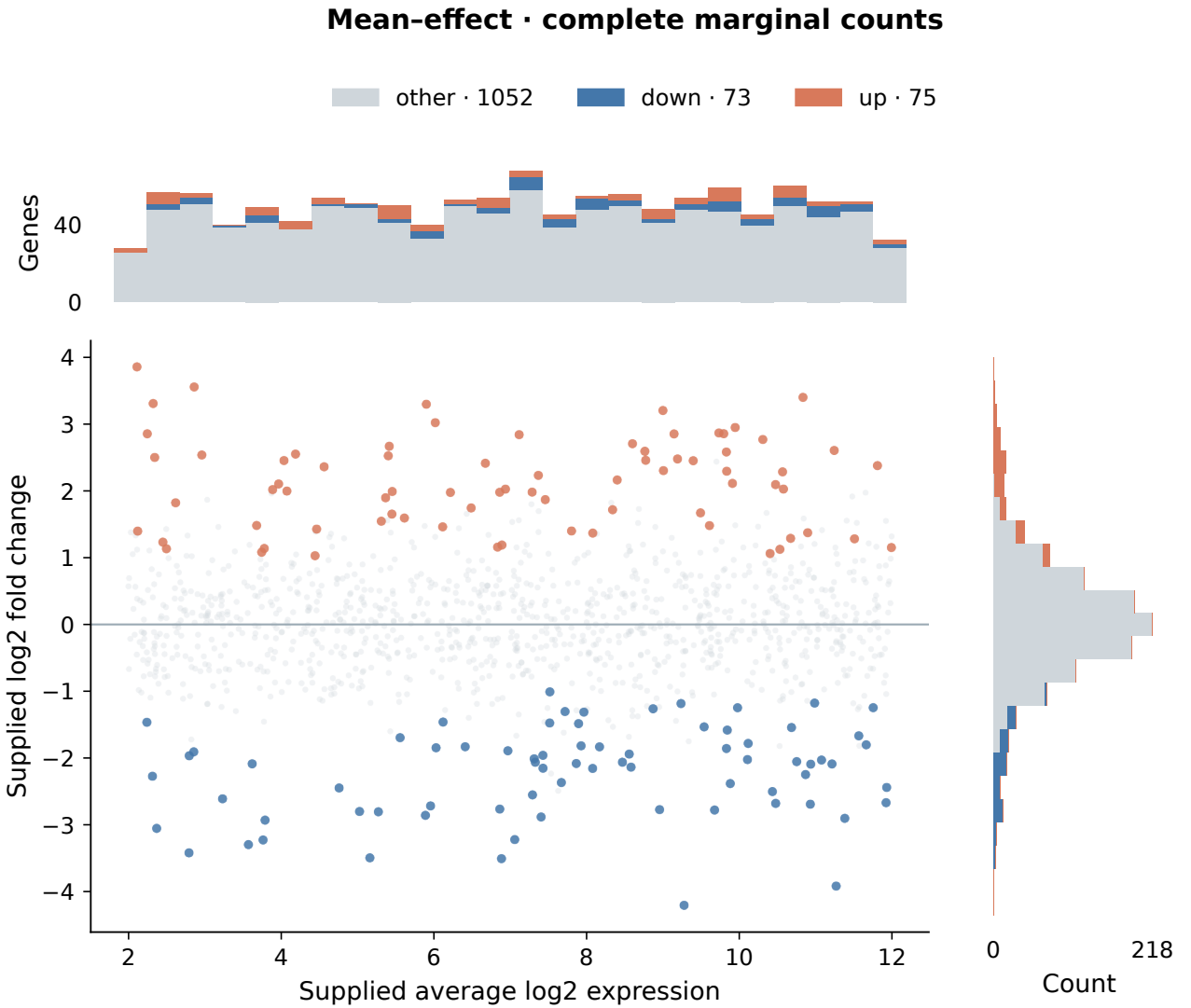
Bulk selected expression programs



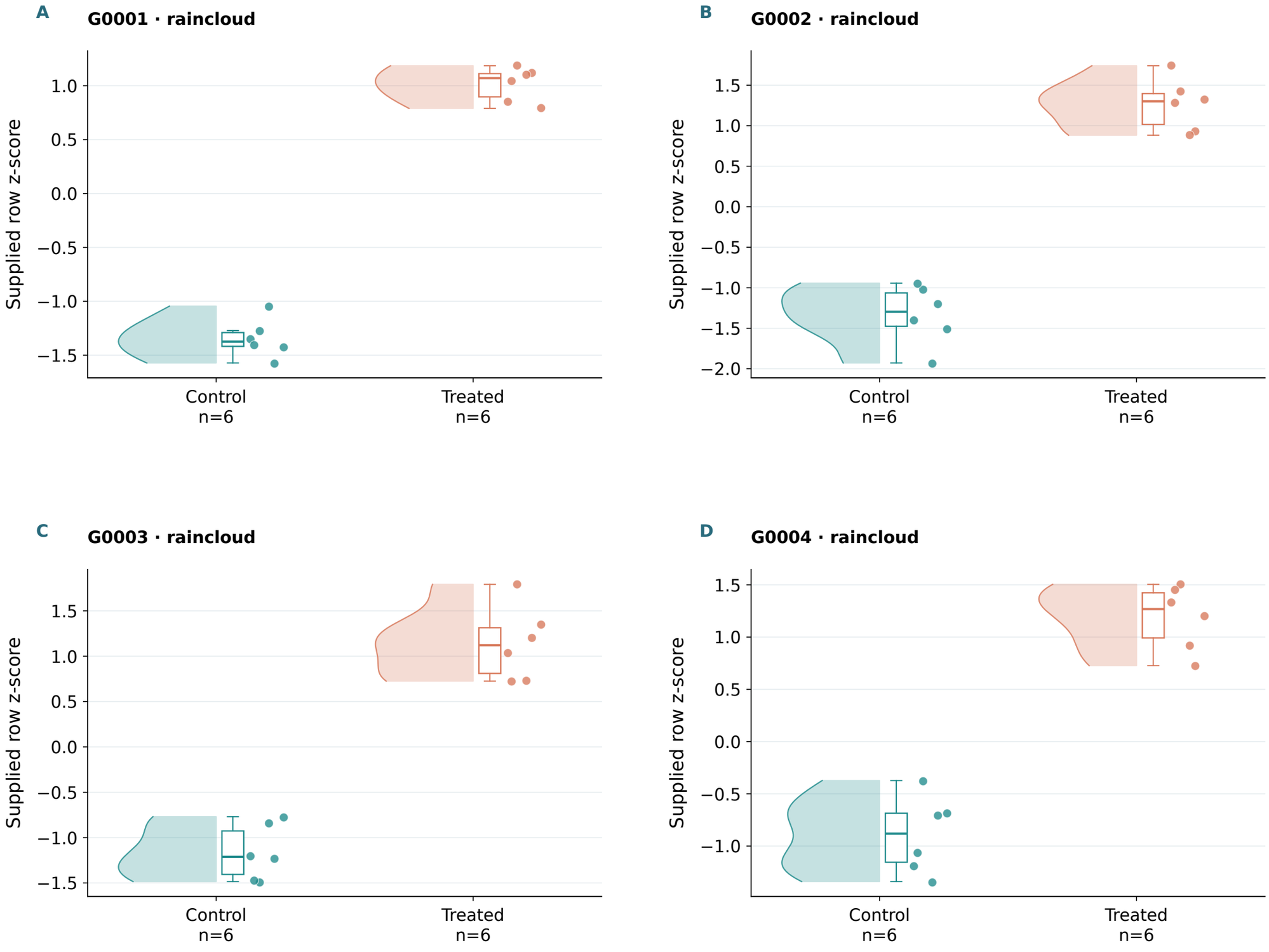
Selected gene effects

Treated versus Control / donor-level



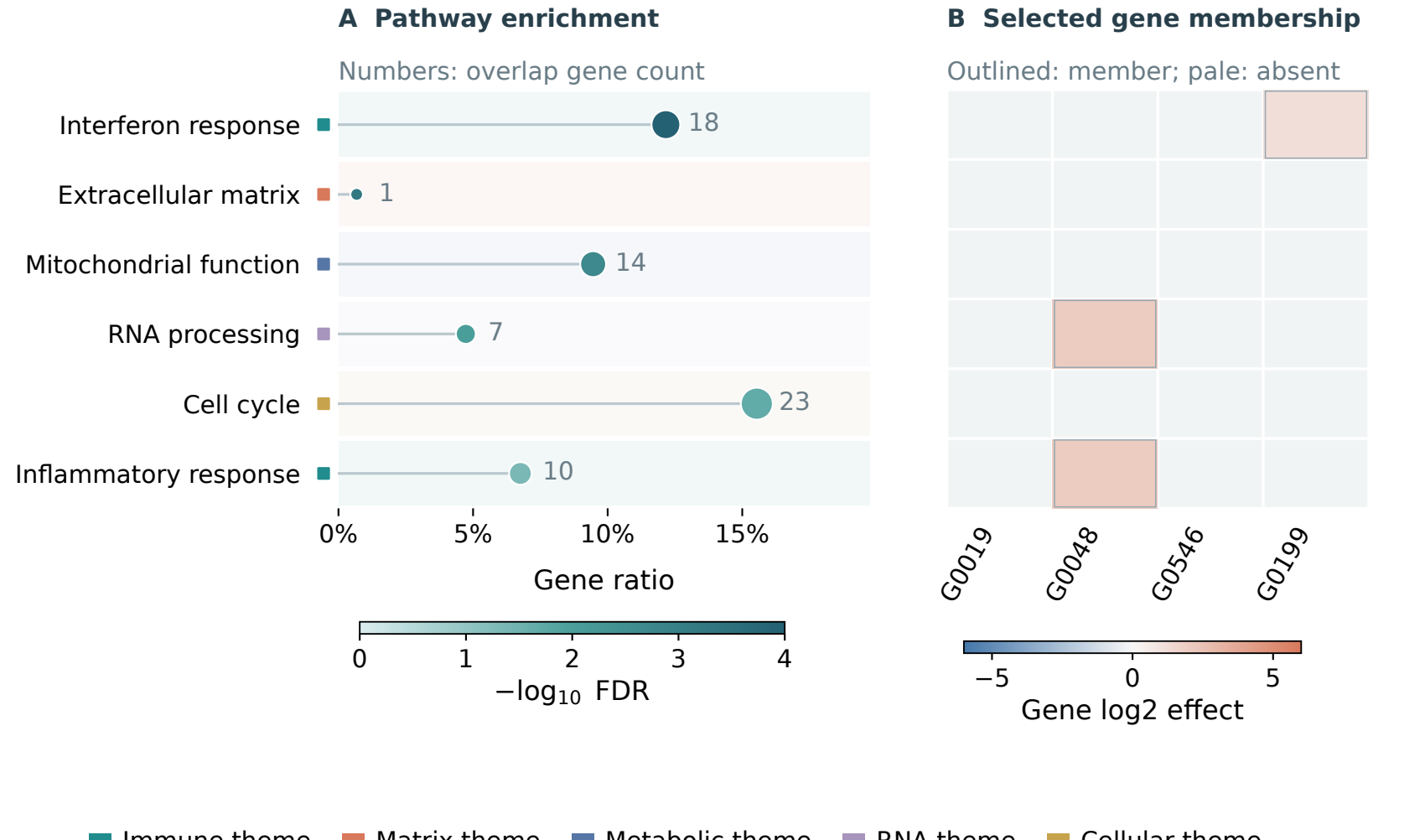

Mean–effect · complete marginal counts
other · 1052
down · 73
up · 75
Genes
40
0
Supplied log2 fold change
4
3
2
1
0
−1
−2
−3
−4
2
4
6
8
10
12
Supplied average log2 expression
0
218
Count

Prespecified donor expression patterns
12 fictional donors · six per condition · supplied results only



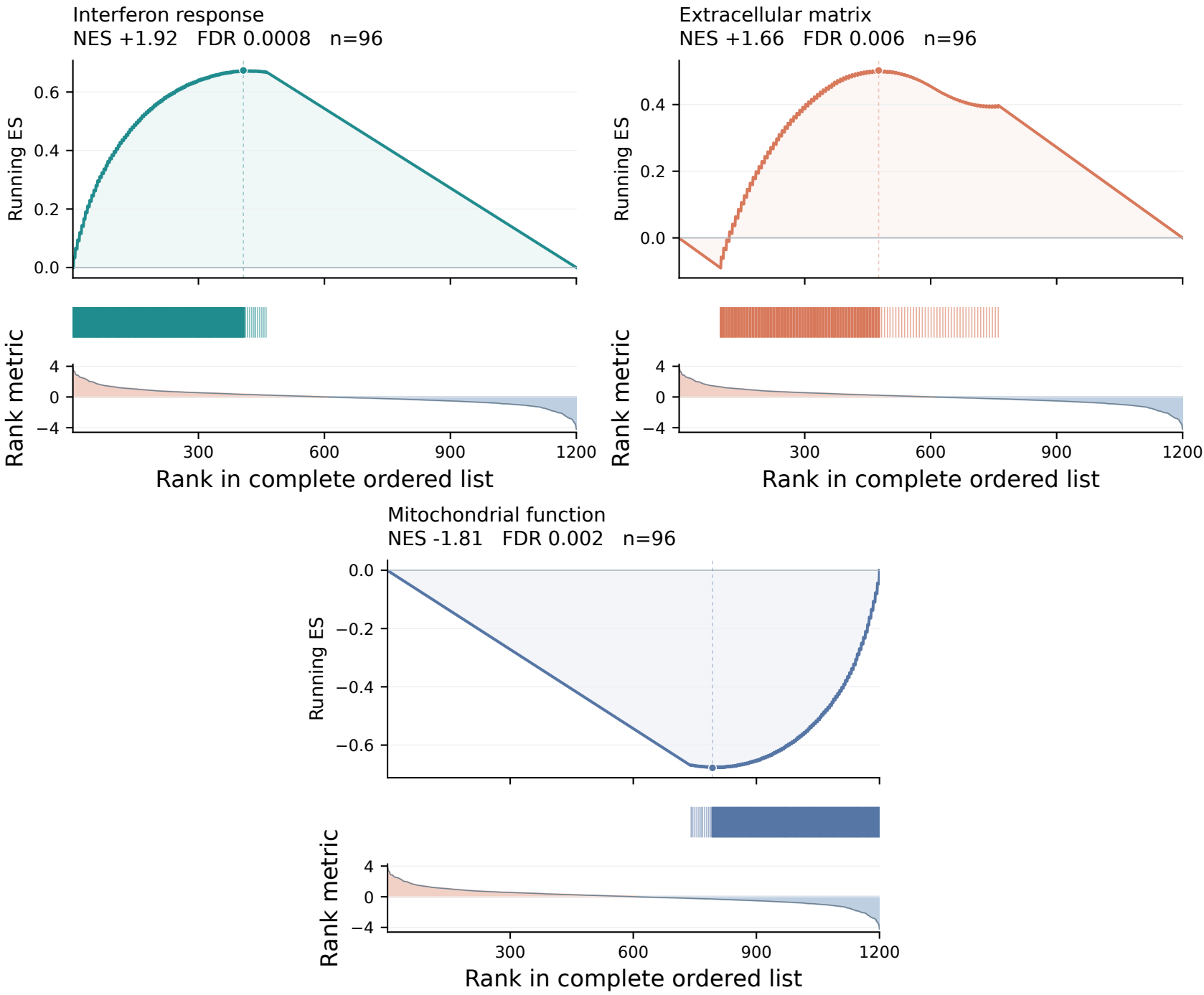
Enrichment themes · ratios, FDR and selected gene support

Gene effects: Treated versus Control / query: 148 / tested: 1200



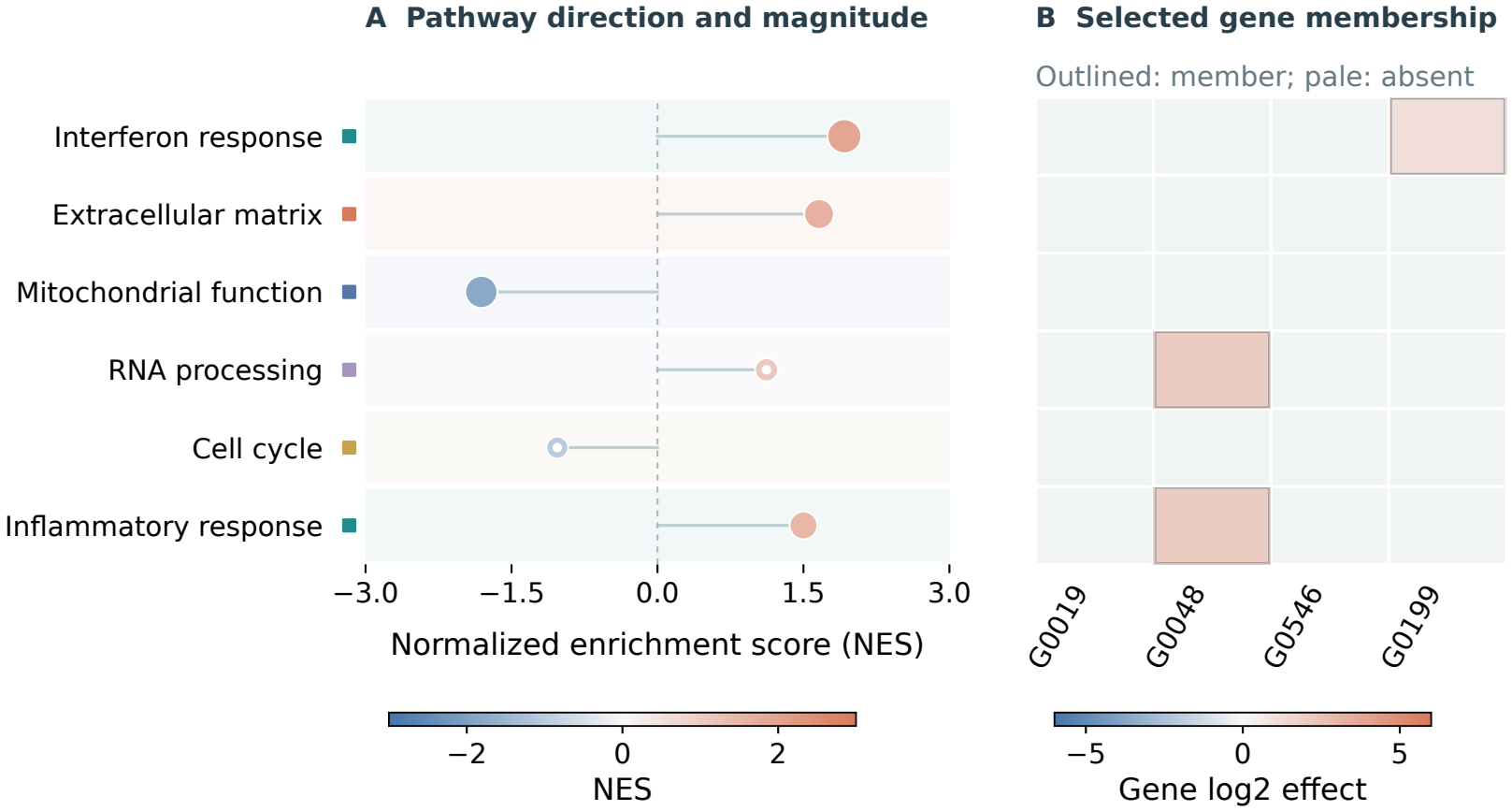
Immune theme | Matrix theme | Metabolic theme | RNA theme | Cellular theme

GSEA · three complete ranked tracks
Treated versus Control / illustrative · Fictional complete
gene set v0.9
Bold hits = supplied leading edge · rank metric: Supplied log2 fold change



GSEA pathway responses and gene support

Gene effects: Treated versus Control / ranked universe: 1200



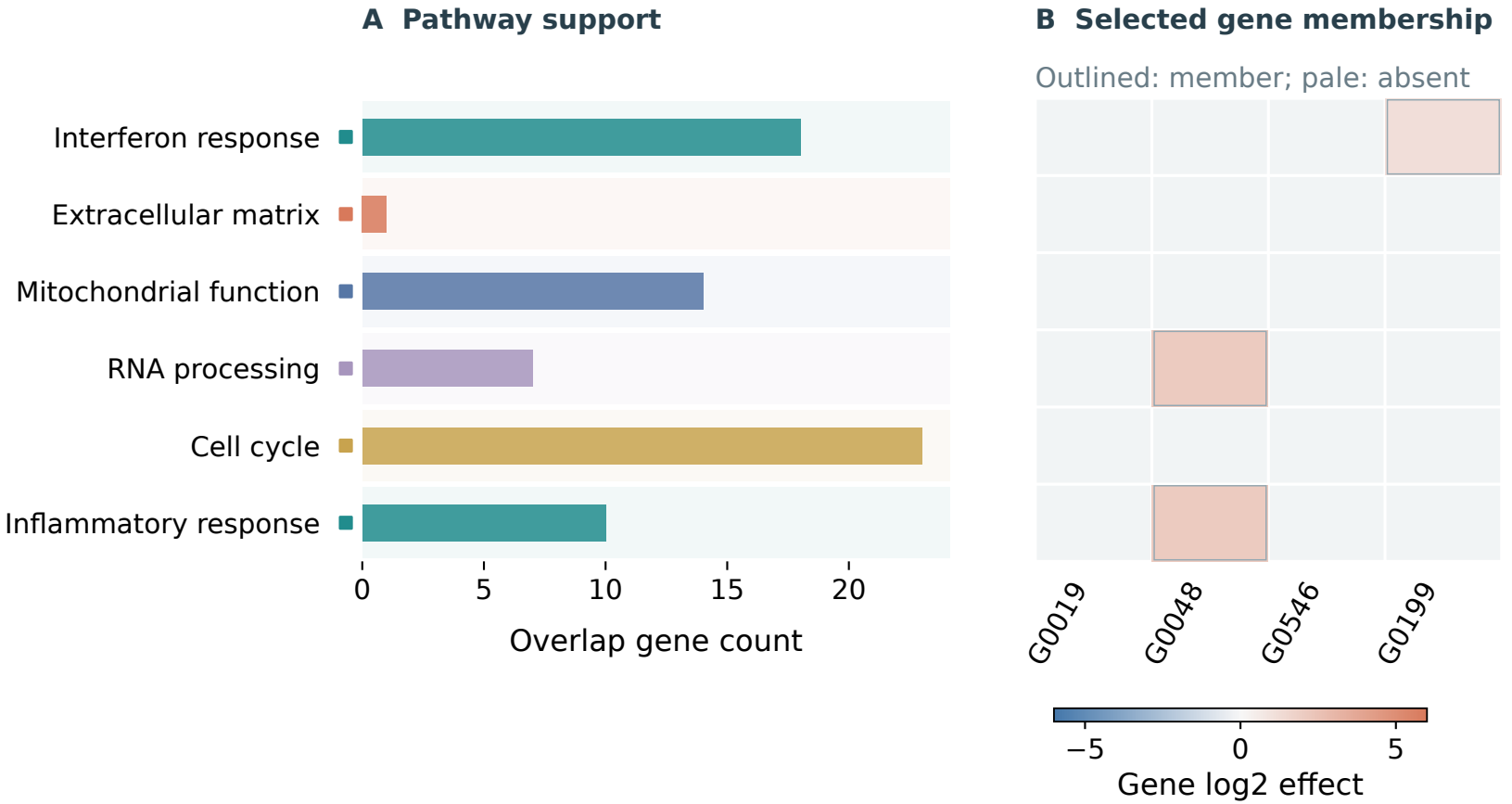
Dot area: -log10 FDR; white centre: FDR > 0.05

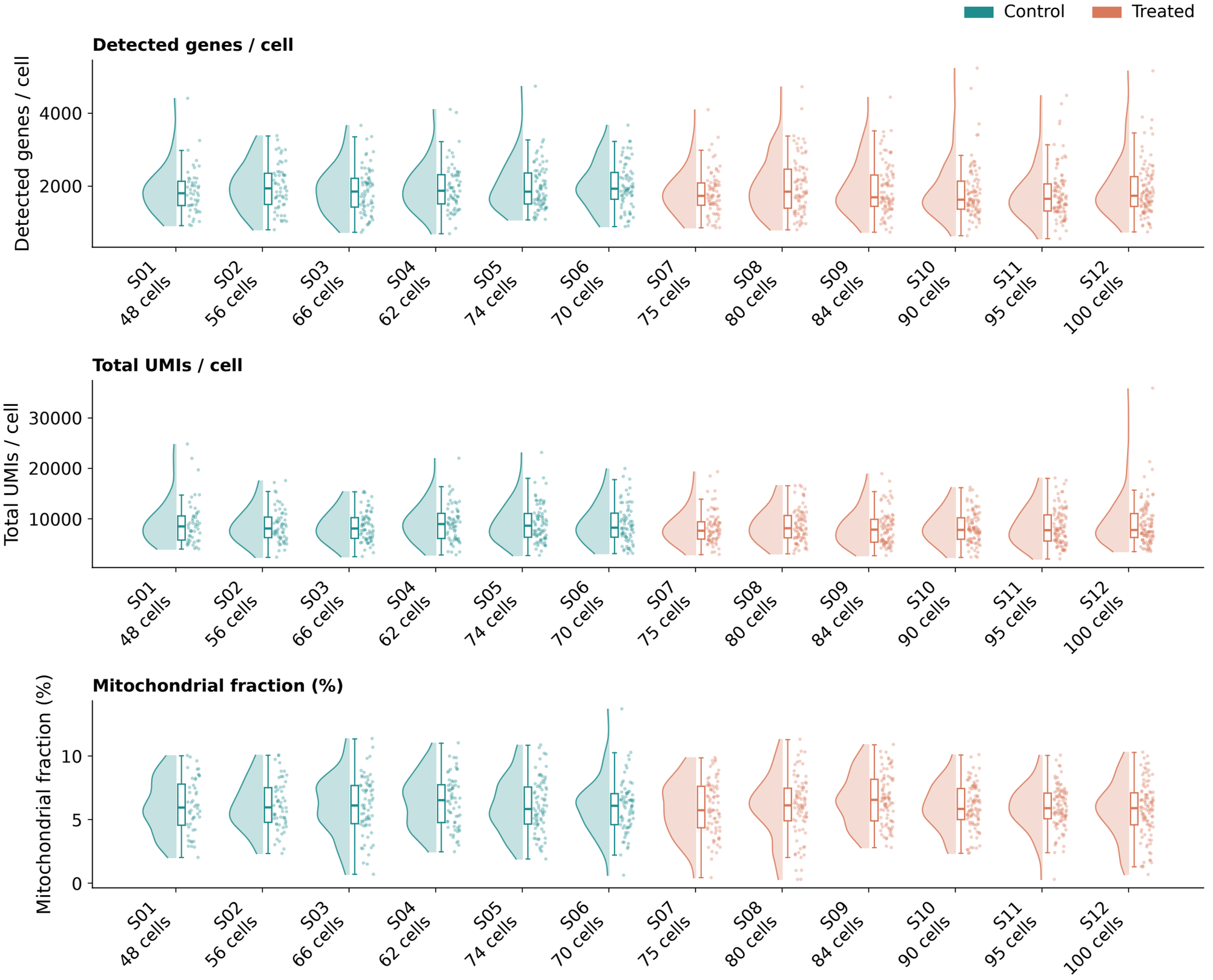
1 3

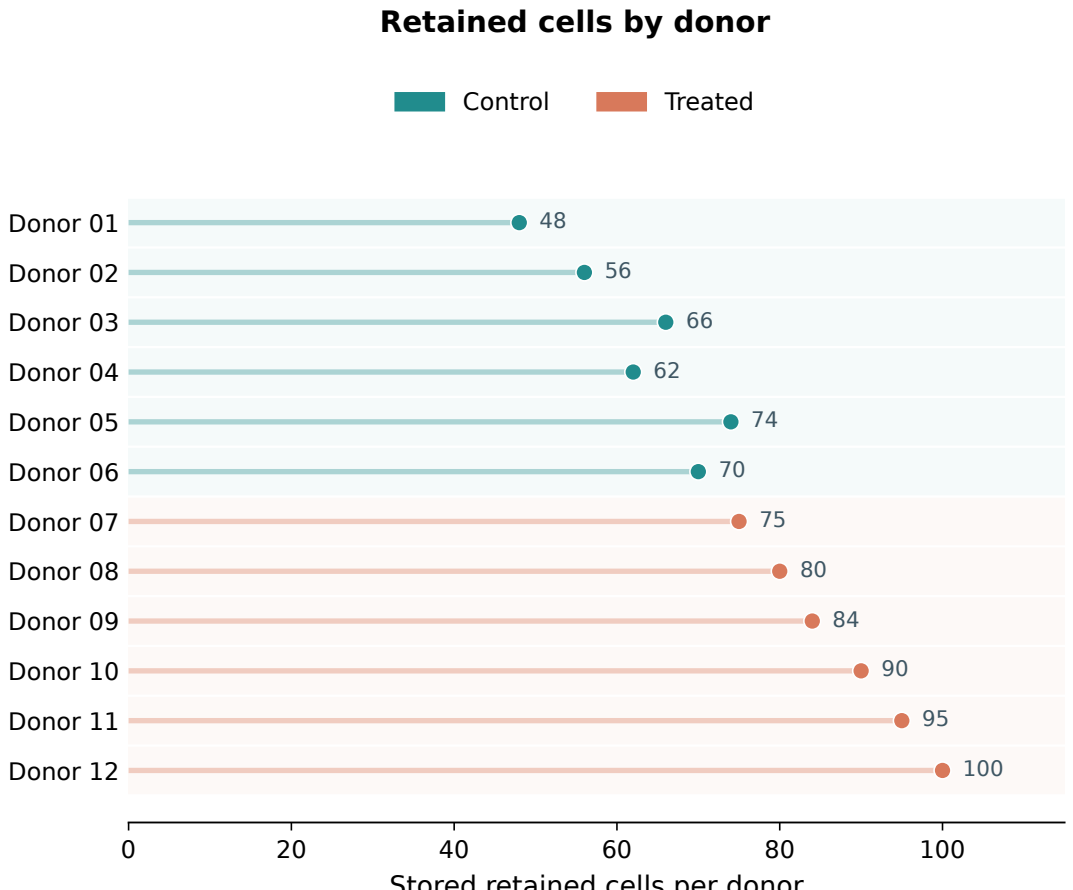
Immune theme Matrix theme Metabolic theme RNA theme Cellular theme

ORA membership and selected gene effects

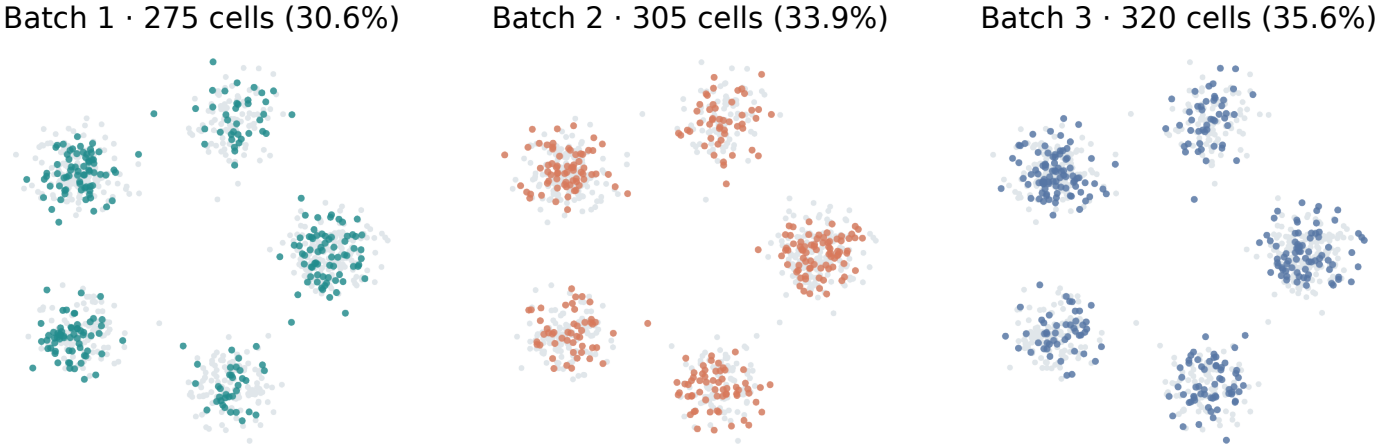
Gene effects: Treated versus Control / query: 148 / tested: 1200




Control
Treated
Detected genes / cell
Detected genes / cell
4000
2000
Total UMIs / cell
Total UMIs / cell
30000
20000
10000
Mitochondrial fraction (%)
Mitochondrial fraction (%)
10
5
0
S01
48 cells
S02
56 cells
S03
66 cells
S04
62 cells
S05
74 cells
S06
70 cells
S07
75 cells
S08
80 cells
S09
84 cells
S10
90 cells
S11
95 cells
S12
100 cells


Retained cells by donor
Control
Treated
Donor 01
48
Donor 02
56
Donor 03
66
Donor 04
62
Donor 05
74
Donor 06
70
Donor 07
75
Donor 08
80
Donor 09
84
Donor 10
90
Donor 11
95
Donor 12
100
0
20
40
60
80
100
Stored retained cells per donor

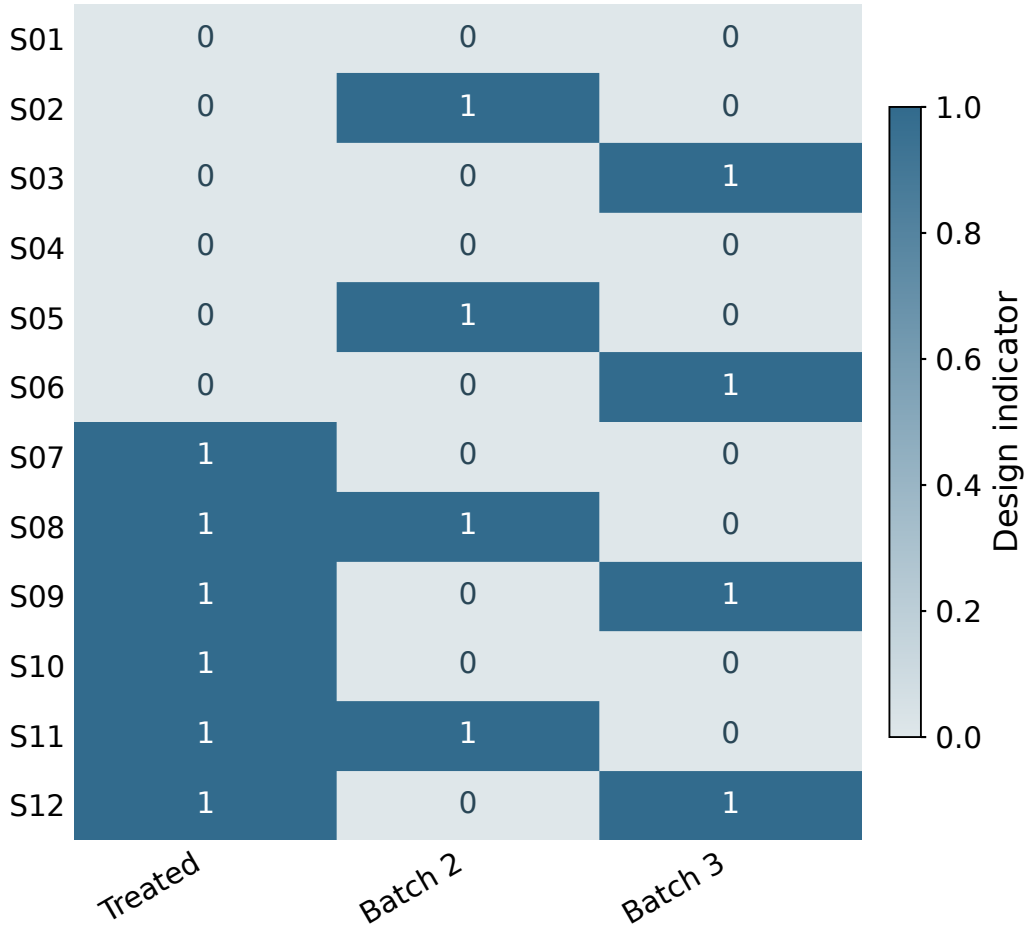
Batch coverage · identical embedding coordinates



Donors, condition and batch
12 fictional donors · six per condition · supplied results only

A

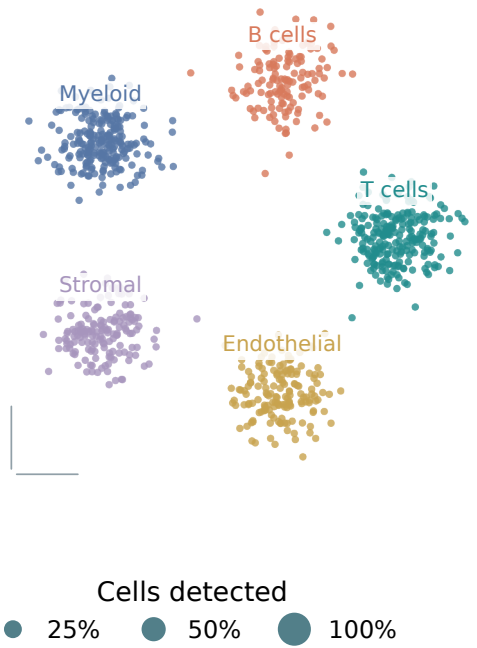
Declared design columns



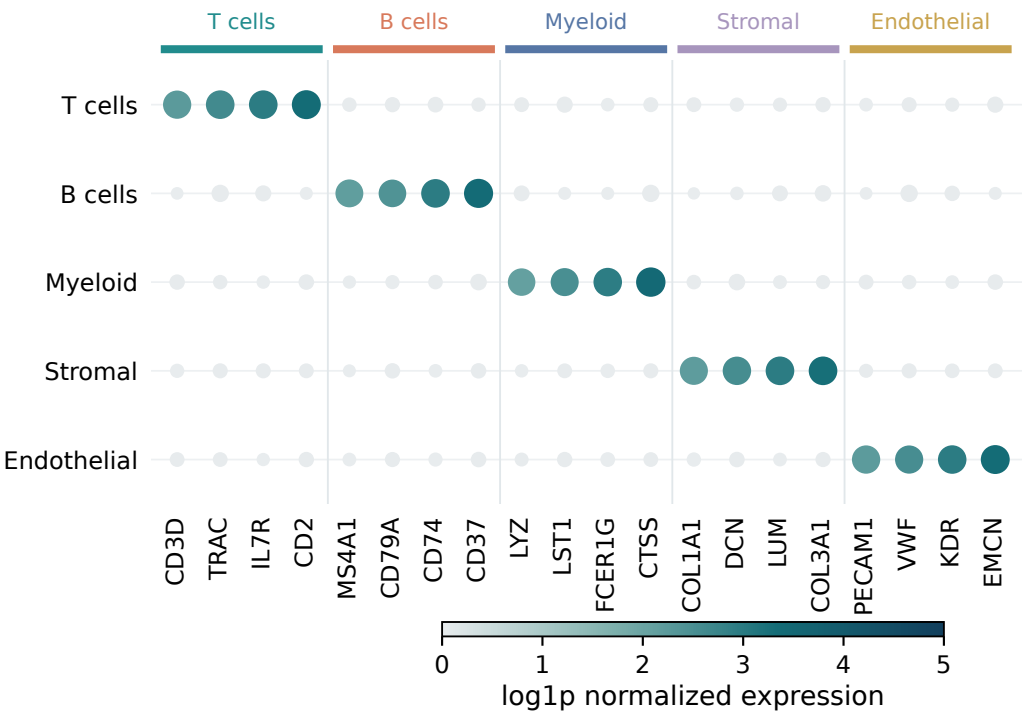
Cell identities and marker evidence

Identity and cross-population marker evidence

A Cell identities

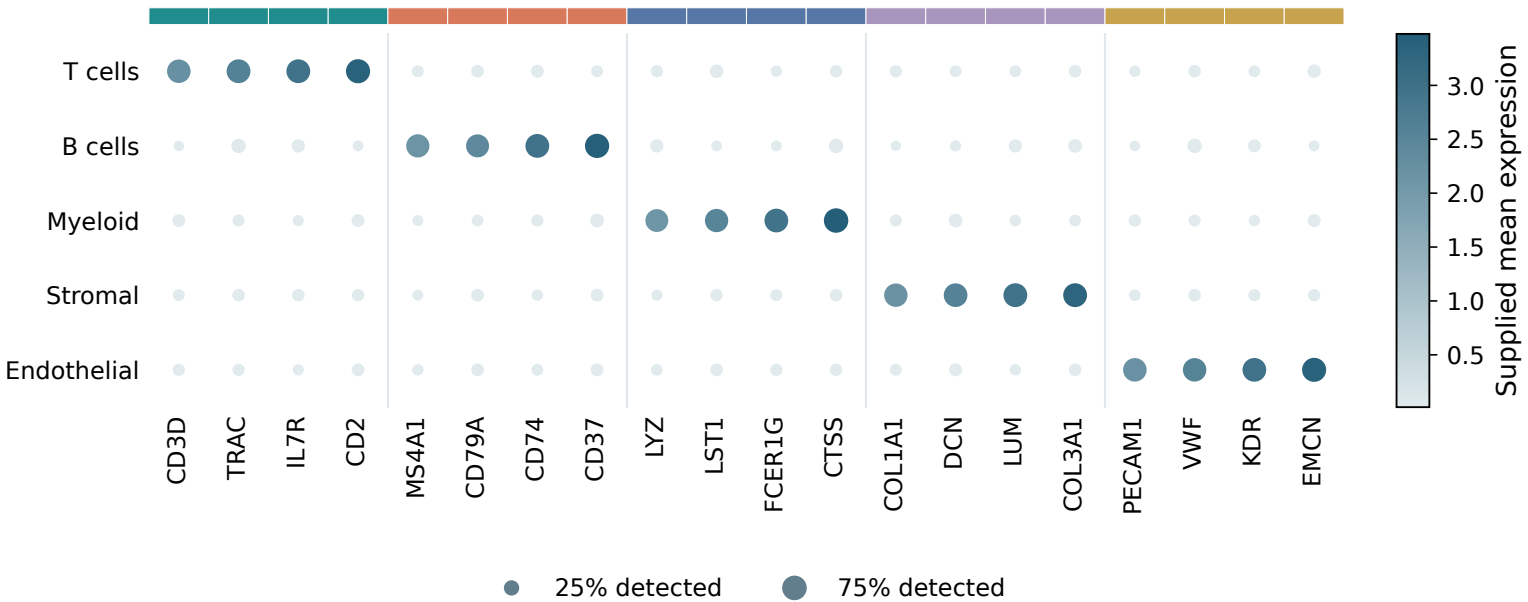


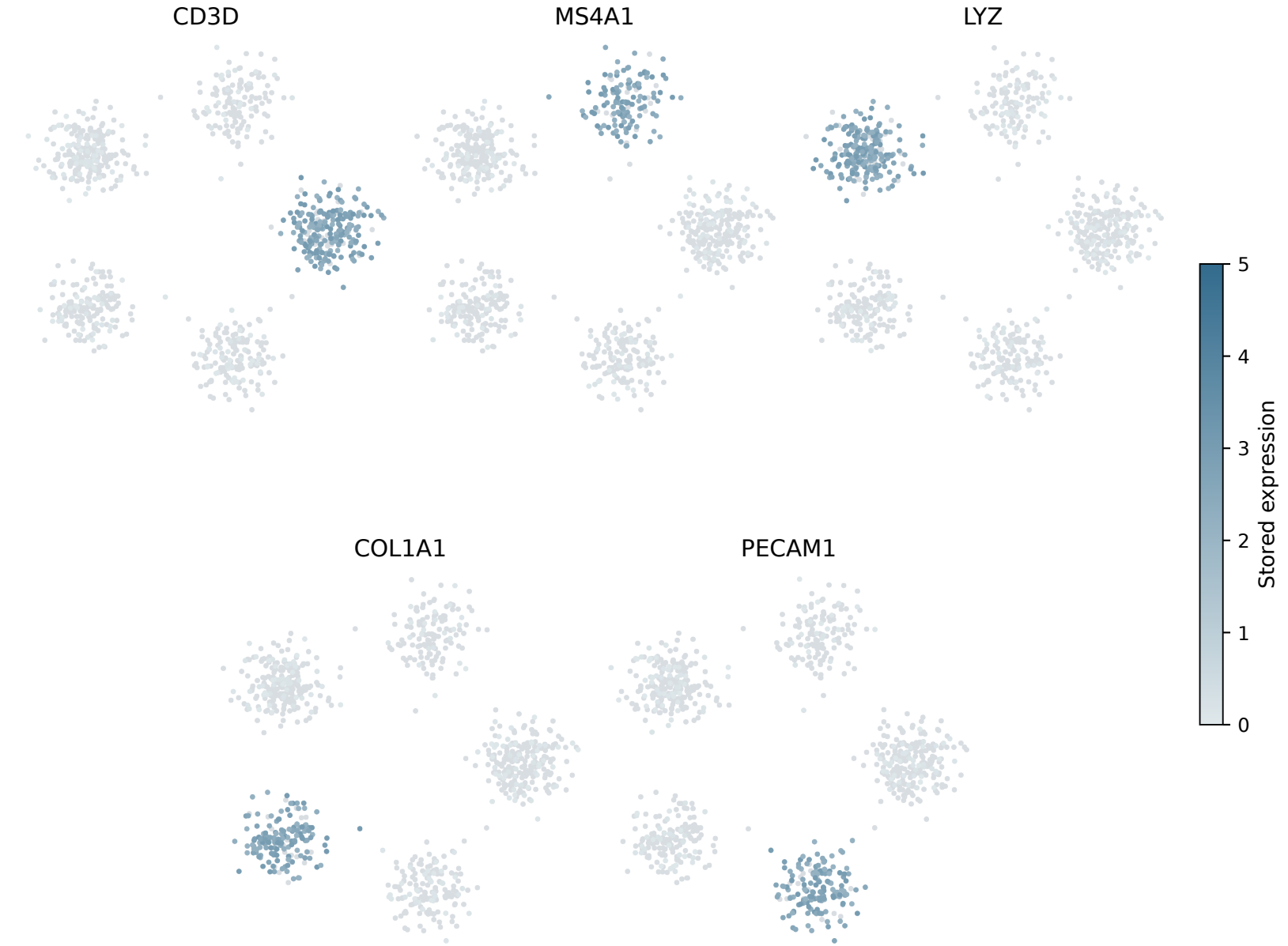
B Marker specificity (mean)



Software test fixture; not biological evidence

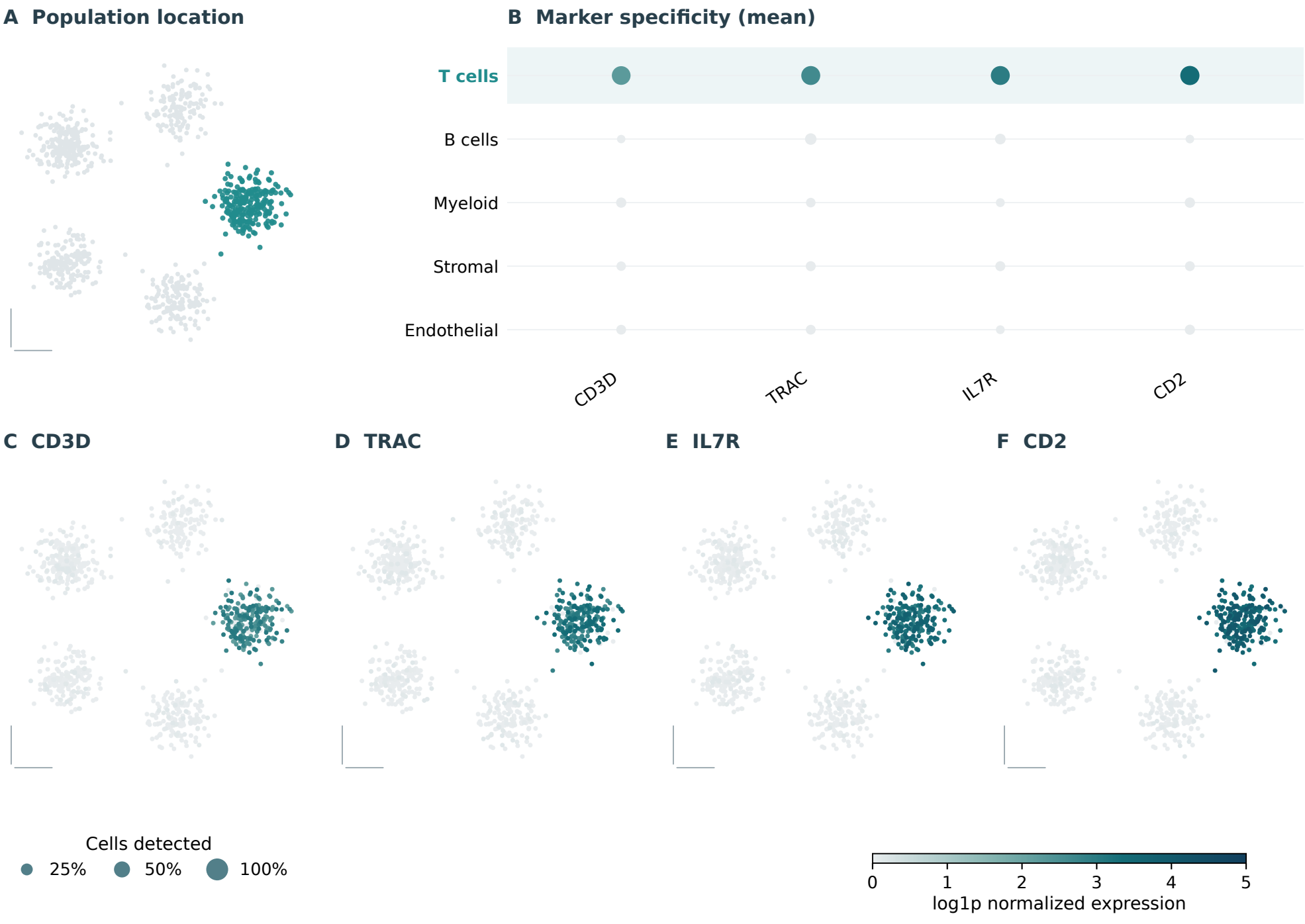
Marker specificity · annotation blocks and detection fractions




CD3D
MS4A1
LYZ
COL1A1
PECAM1
Stored expression
0
1
2
3
4
5

T cells · four-marker evidence

T cells / 226 of 900 displayed cells / 4 selected markers



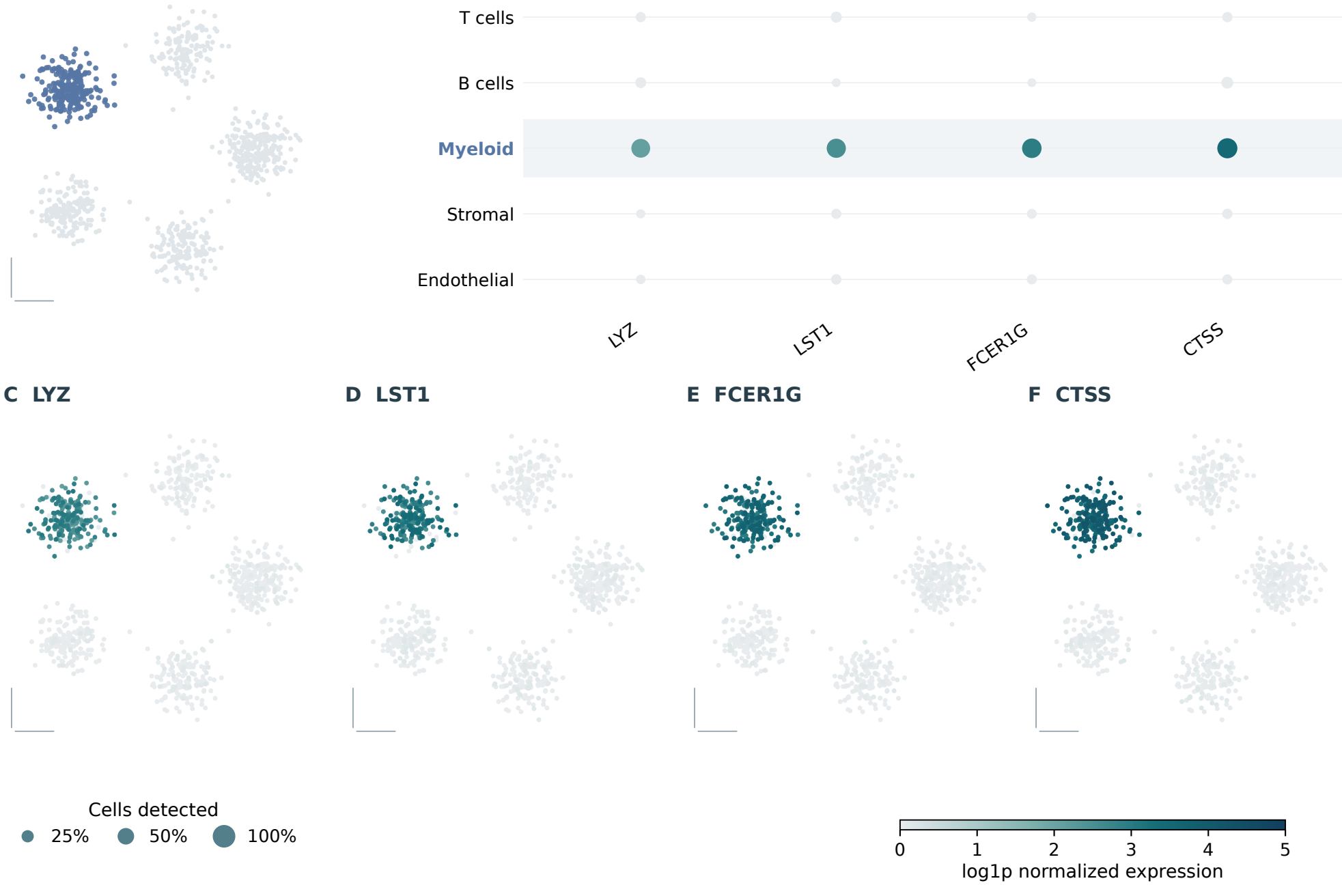
B cells · four-marker evidence

B cells / 132 of 900 displayed cells / 4 selected markers



Myeloid · four-marker evidence

Myeloid / 226 of 900 displayed cells / 4 selected markers



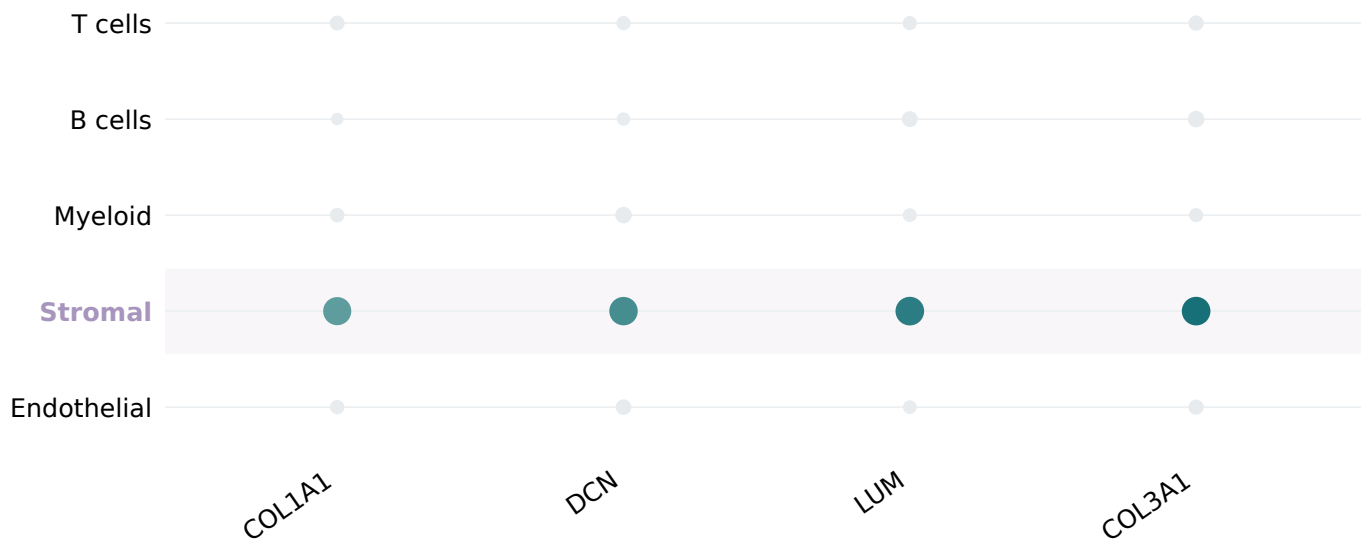
Stromal · four-marker evidence

Stromal / 159 of 900 displayed cells / 4 selected markers

A Population location



B Marker specificity (mean)



C COL1A1



D DCN

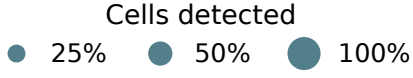


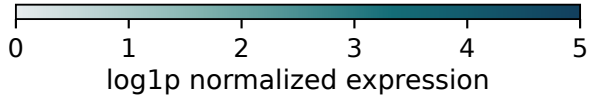
E LUM



F COL3A1



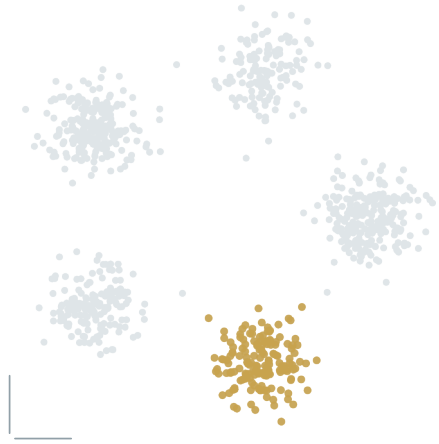




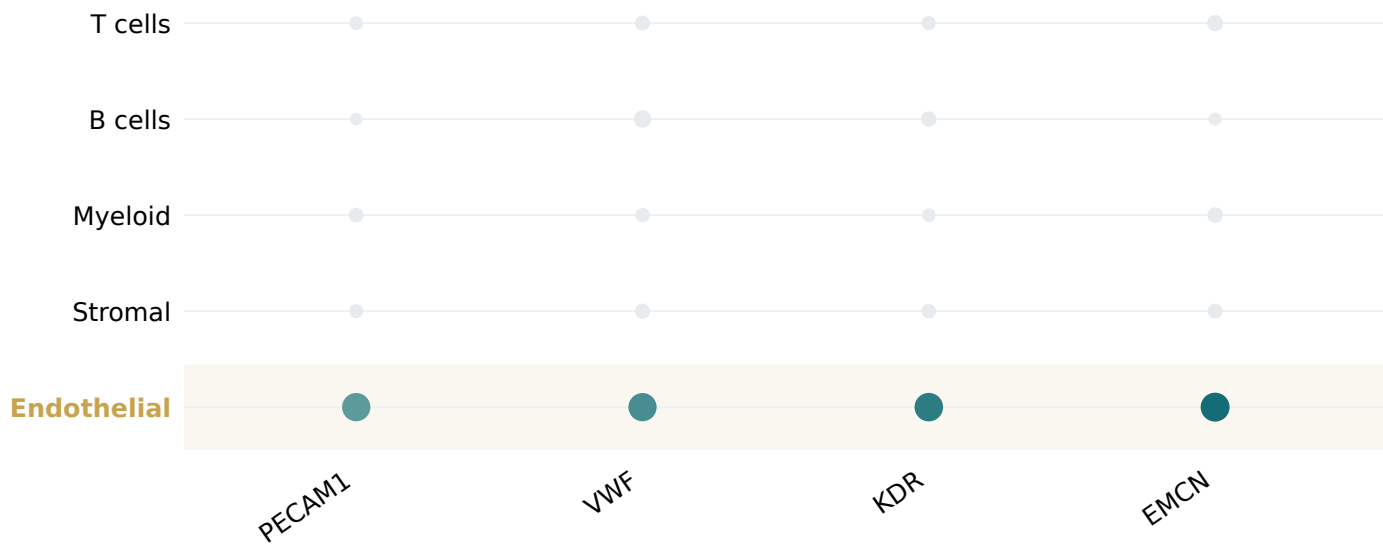
Endothelial · four-marker evidence

Endothelial / 157 of 900 displayed cells / 4 selected markers

A Population location



B Marker specificity (mean)



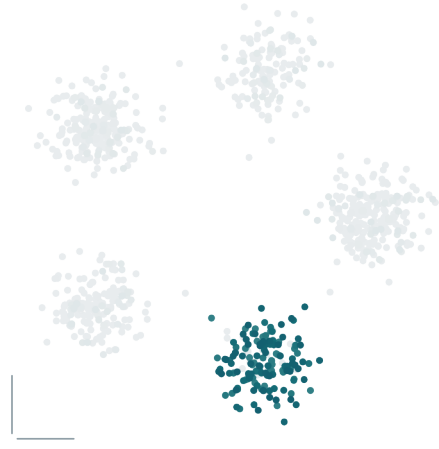
C PECAM1



D VWF



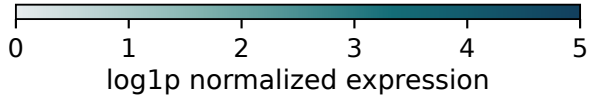
E KDR



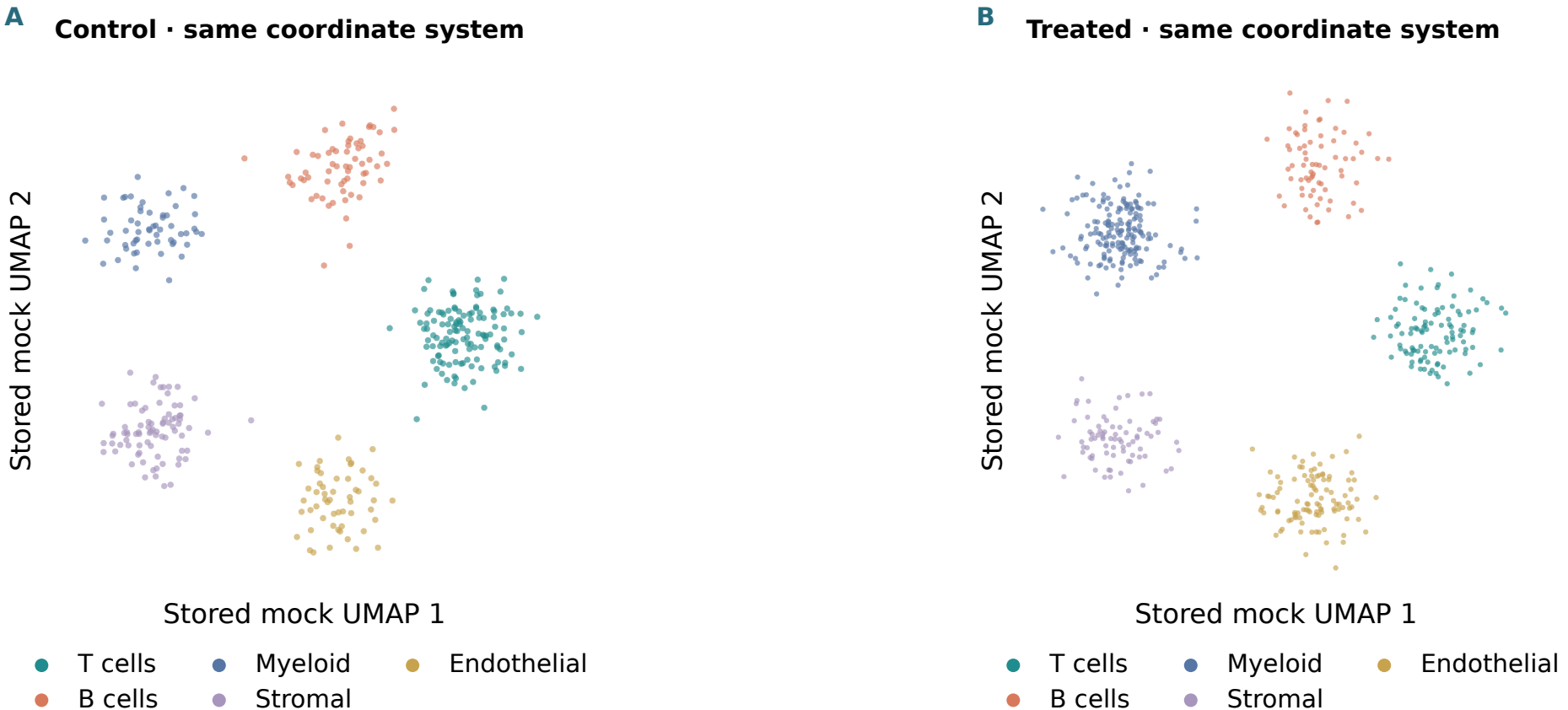
F EMCN

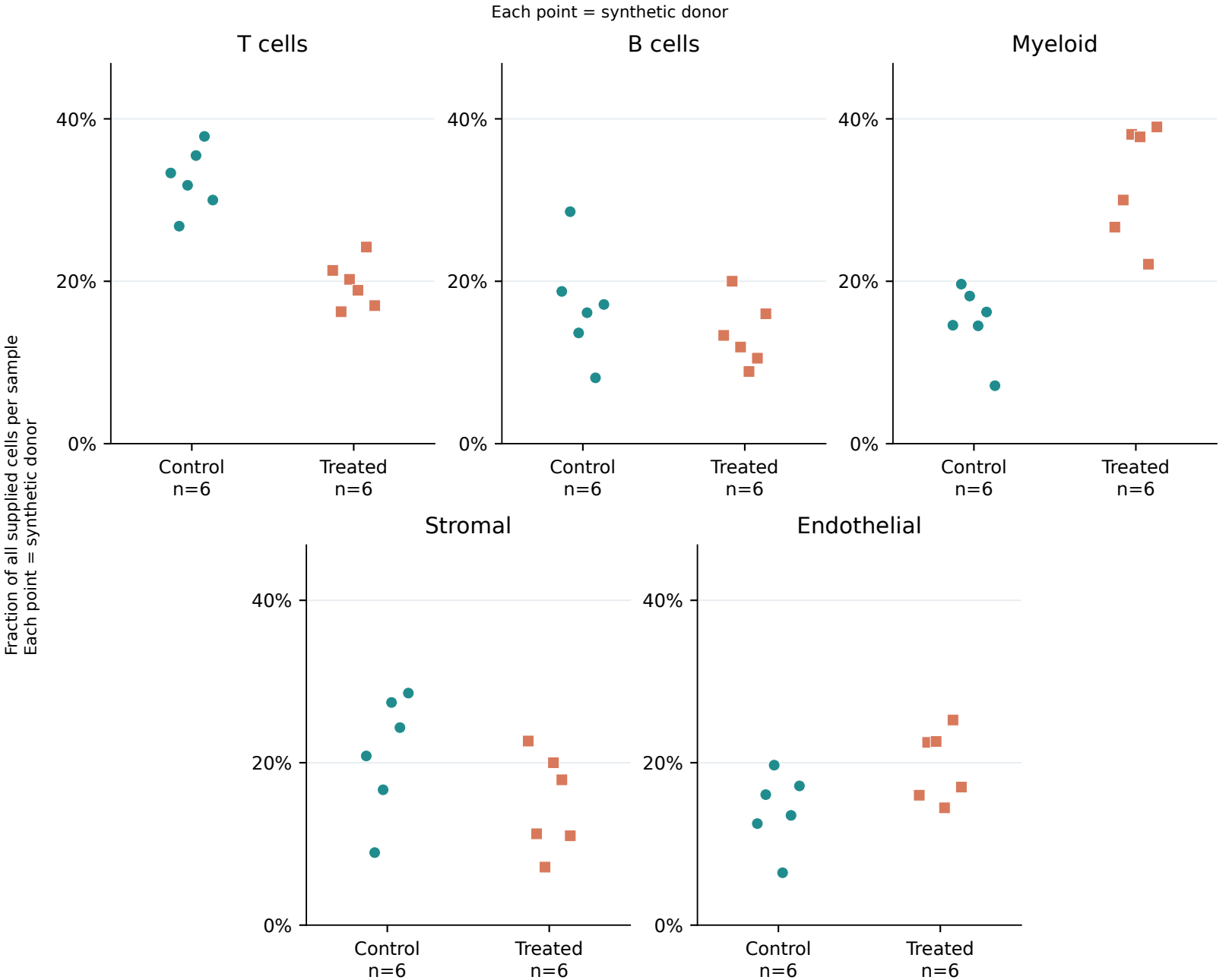






Cell identities by condition
12 fictional donors · six per condition · supplied results only

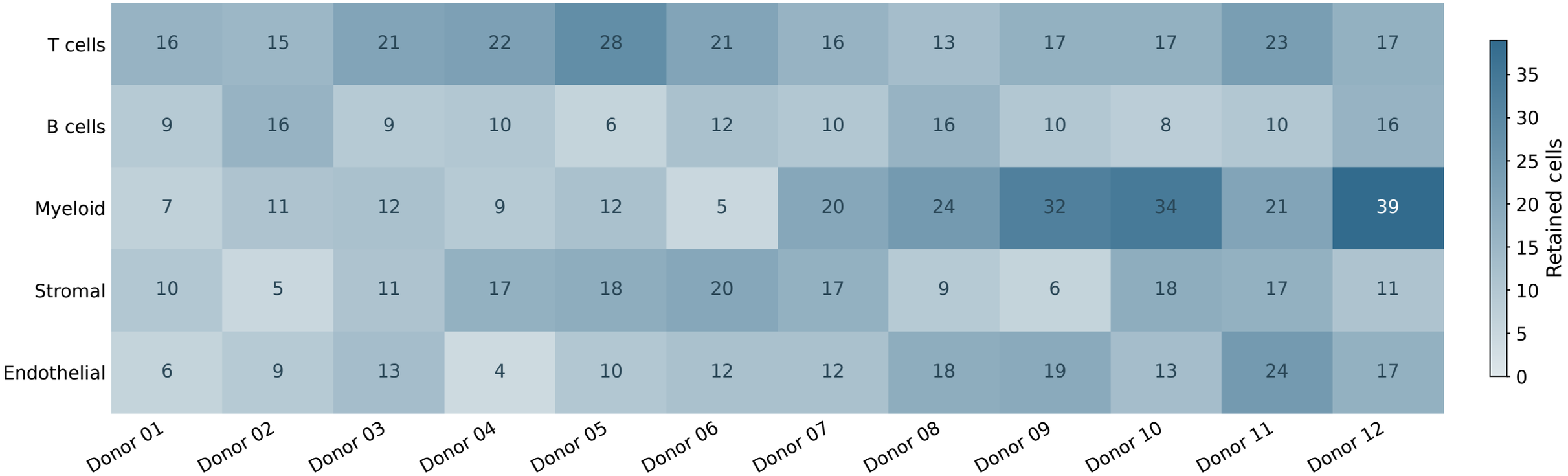


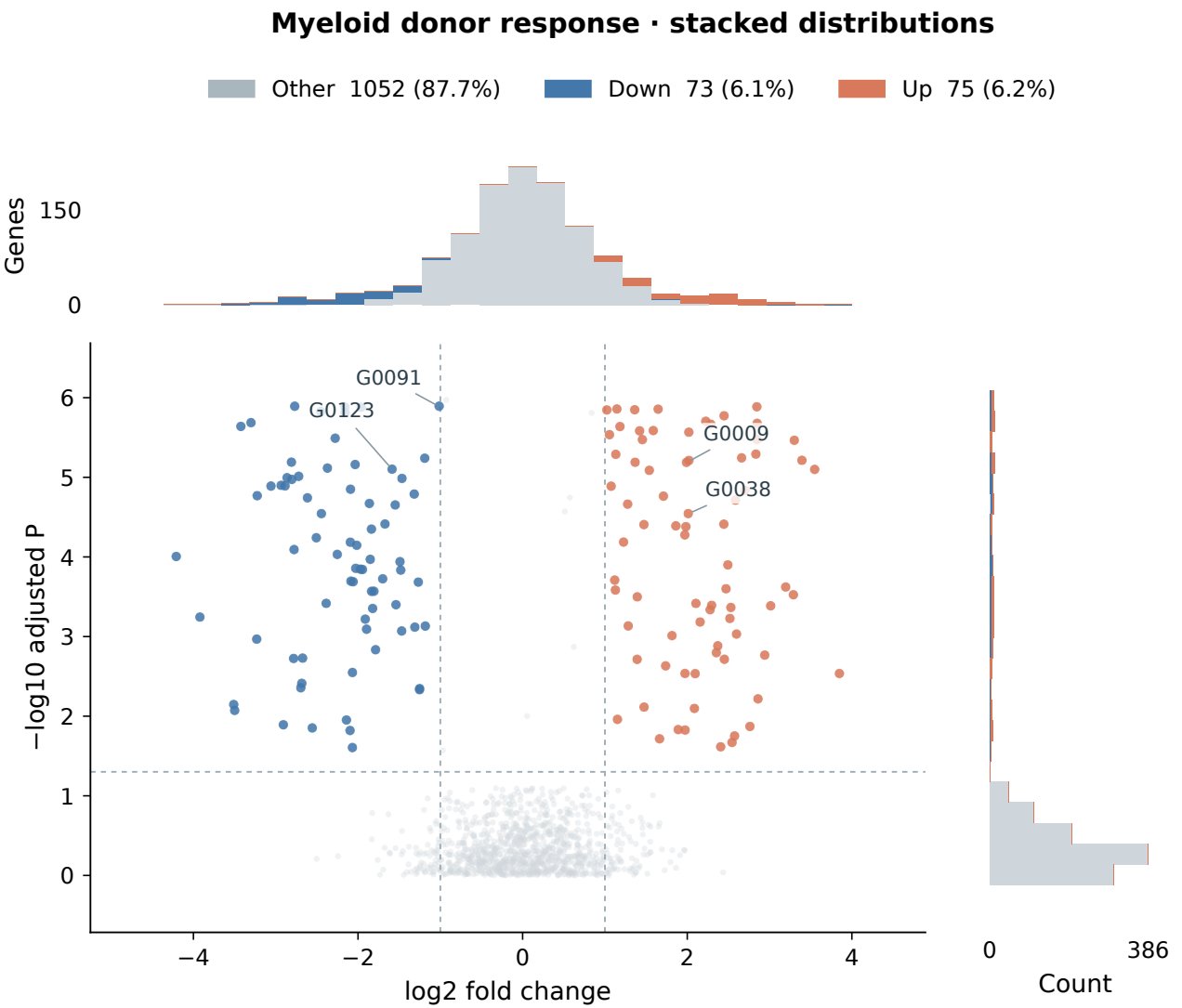

Each point = synthetic donor
T cells
B cells
Myeloid
Stromal
Endothelial
Fraction of all supplied cells per sample
Each point = synthetic donor
40%
20%
0%
Control
n=6
Treated
n=6

Cell support and donor coverage
12 fictional donors · six per condition · supplied results only

A

Independent donor support for every identity



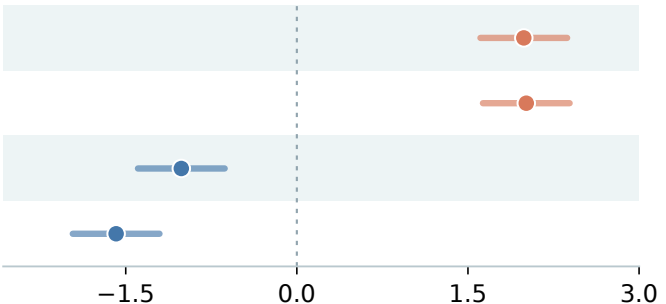


Stacked marginals retain all genes · FDR ≤ 0.05 and |log2FC| ≥ 1

Myeloid donor effects

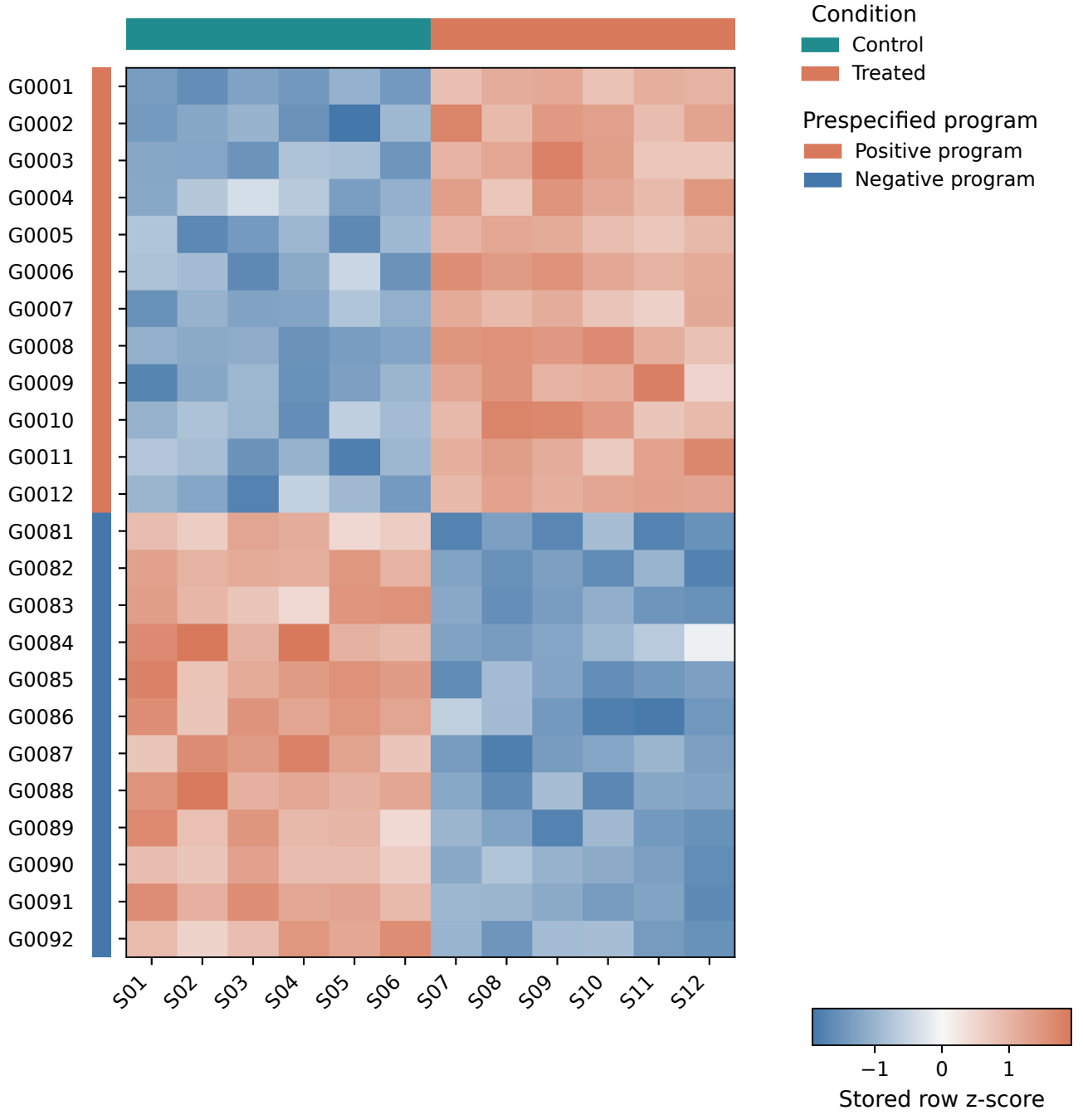
Treated versus Control / Myeloid donor-level

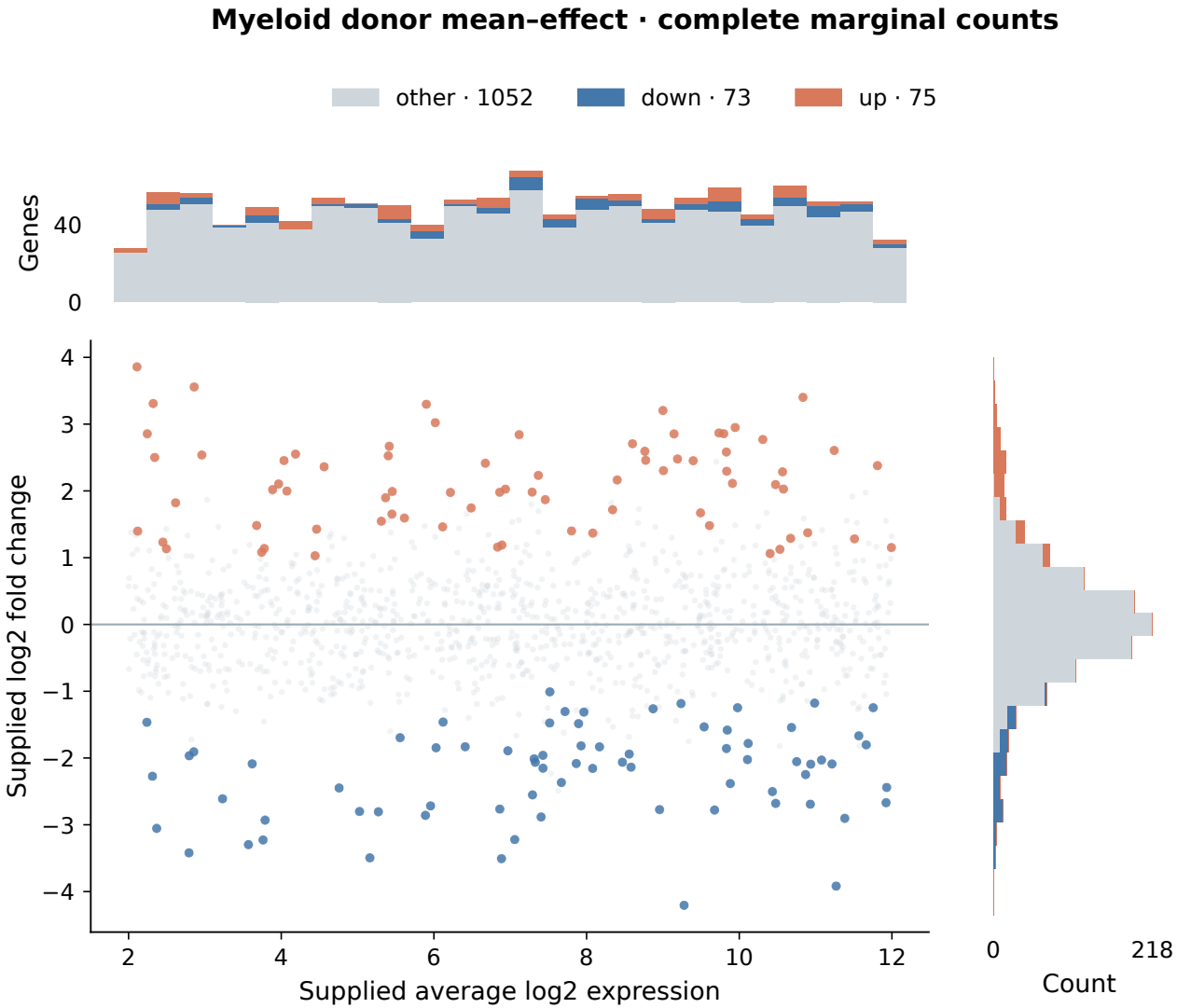
Gene	log2 fold change	Estimate [95% interval]
G0009		+1.99 [+1.61, +2.37]
G0038		+2.01 [+1.63, +2.39]
G0091		-1.01 [-1.39, -0.63]
G0123		-1.58 [-1.96, -1.20]



Points: supplied estimates · Lines: supplied 95% intervals

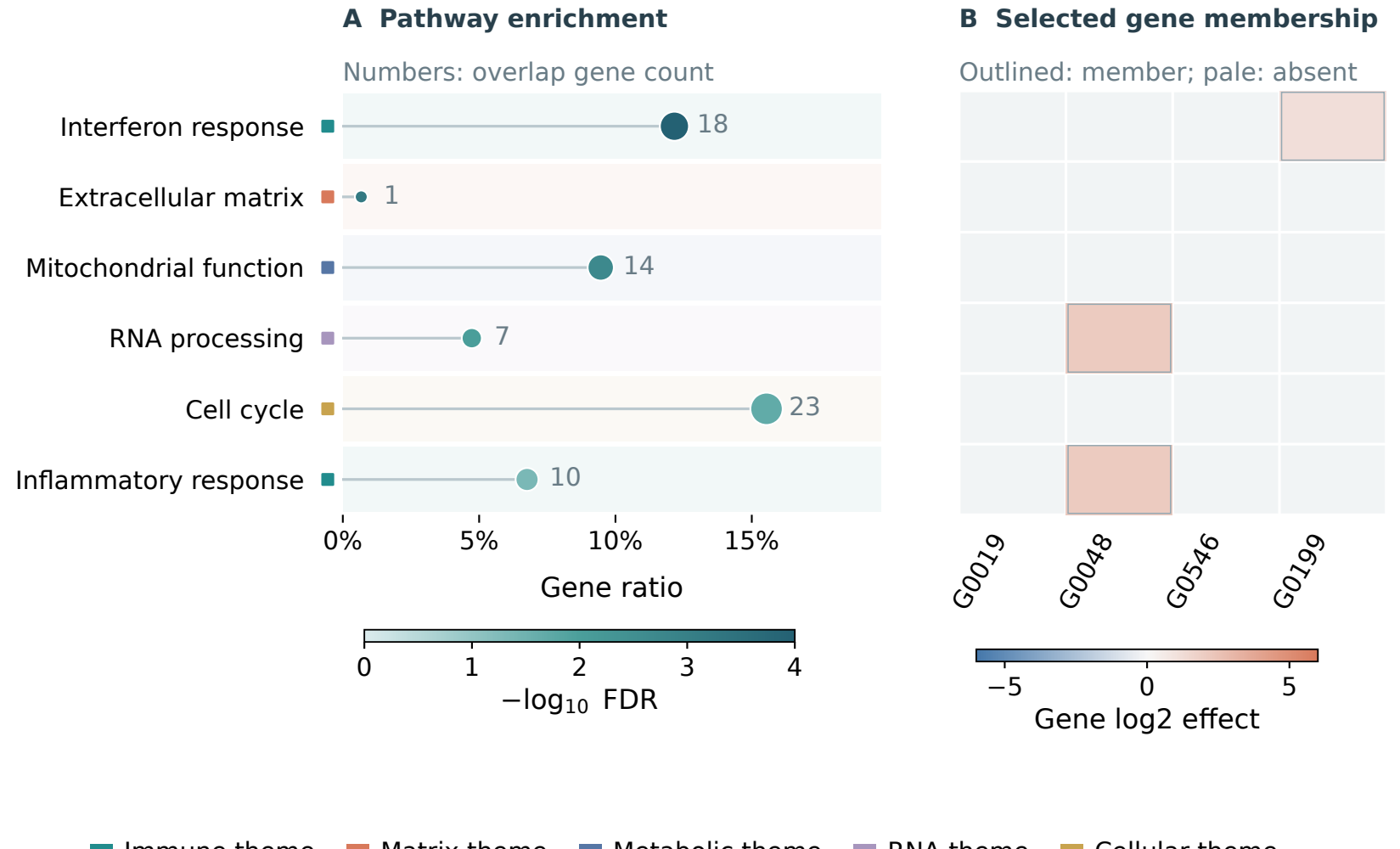
Myeloid donor expression programs




Myeloid donor mean–effect · complete marginal counts
other · 1052
down · 73
up · 75
Genes
40
0
Supplied log2 fold change
4
3
2
1
0
−1
−2
−3
−4
2
4
6
8
10
12
Supplied average log2 expression
0
218
Count

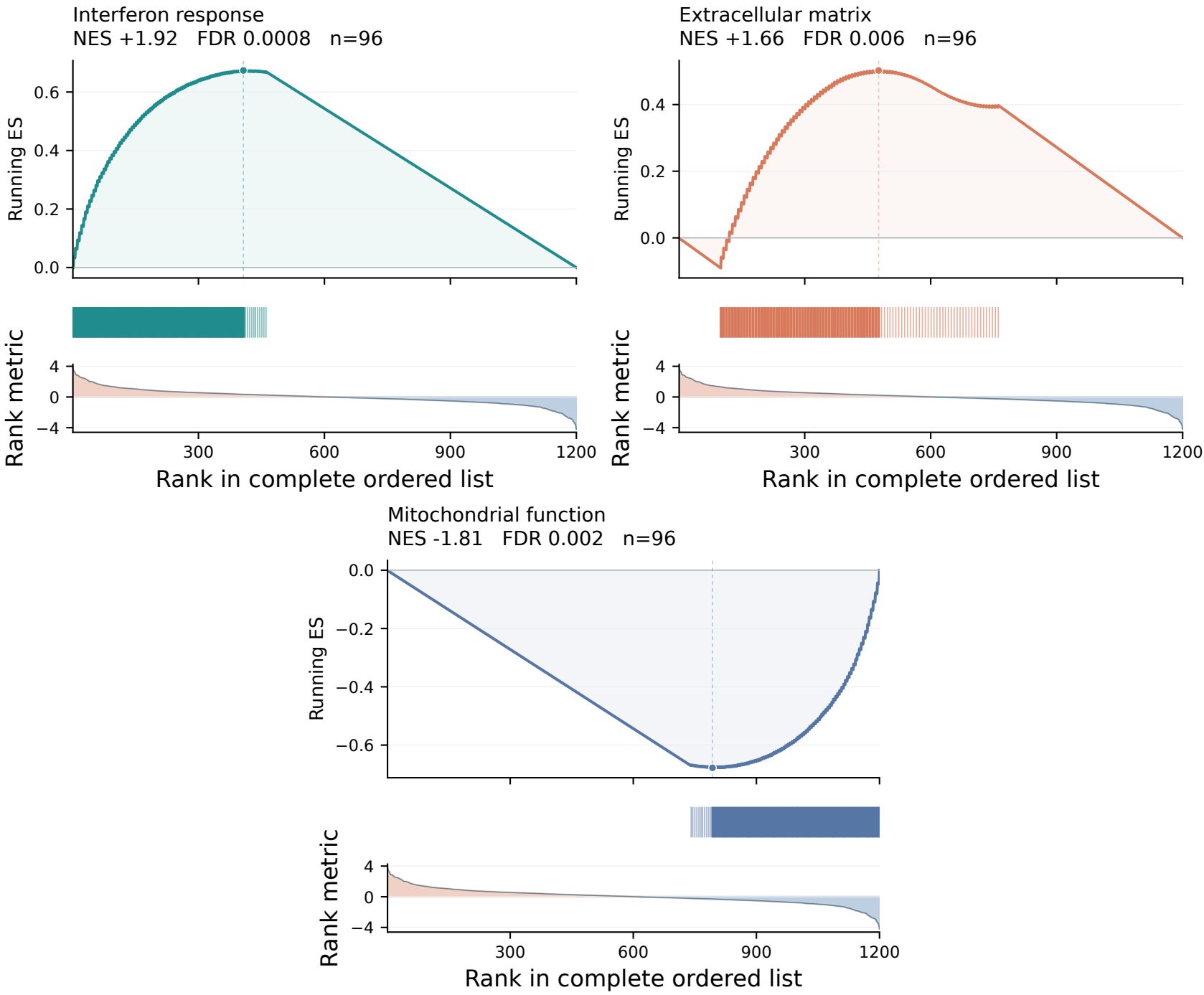
Myeloid · Enrichment themes · ratios, FDR and selected gene support

Gene effects: Treated versus Control / query: 148 / tested: 1200



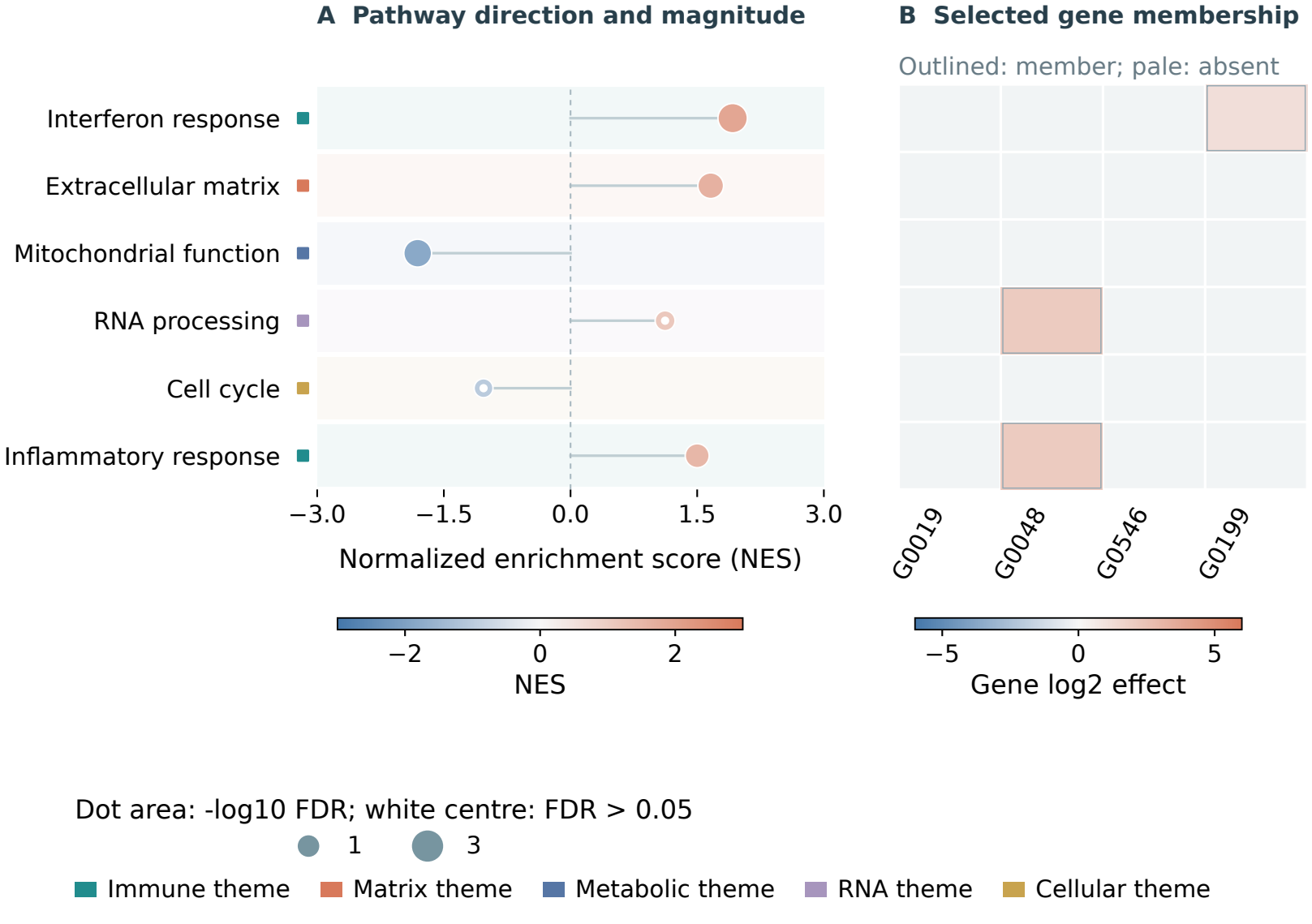
Immune theme · Matrix theme · Metabolic theme · RNA theme · Cellular theme

Myeloid · GSEA · three complete ranked tracks
Treated versus Control / illustrative · Fictional complete gene set v0.9
Bold hits = supplied leading edge · rank metric: Supplied log2 fold change



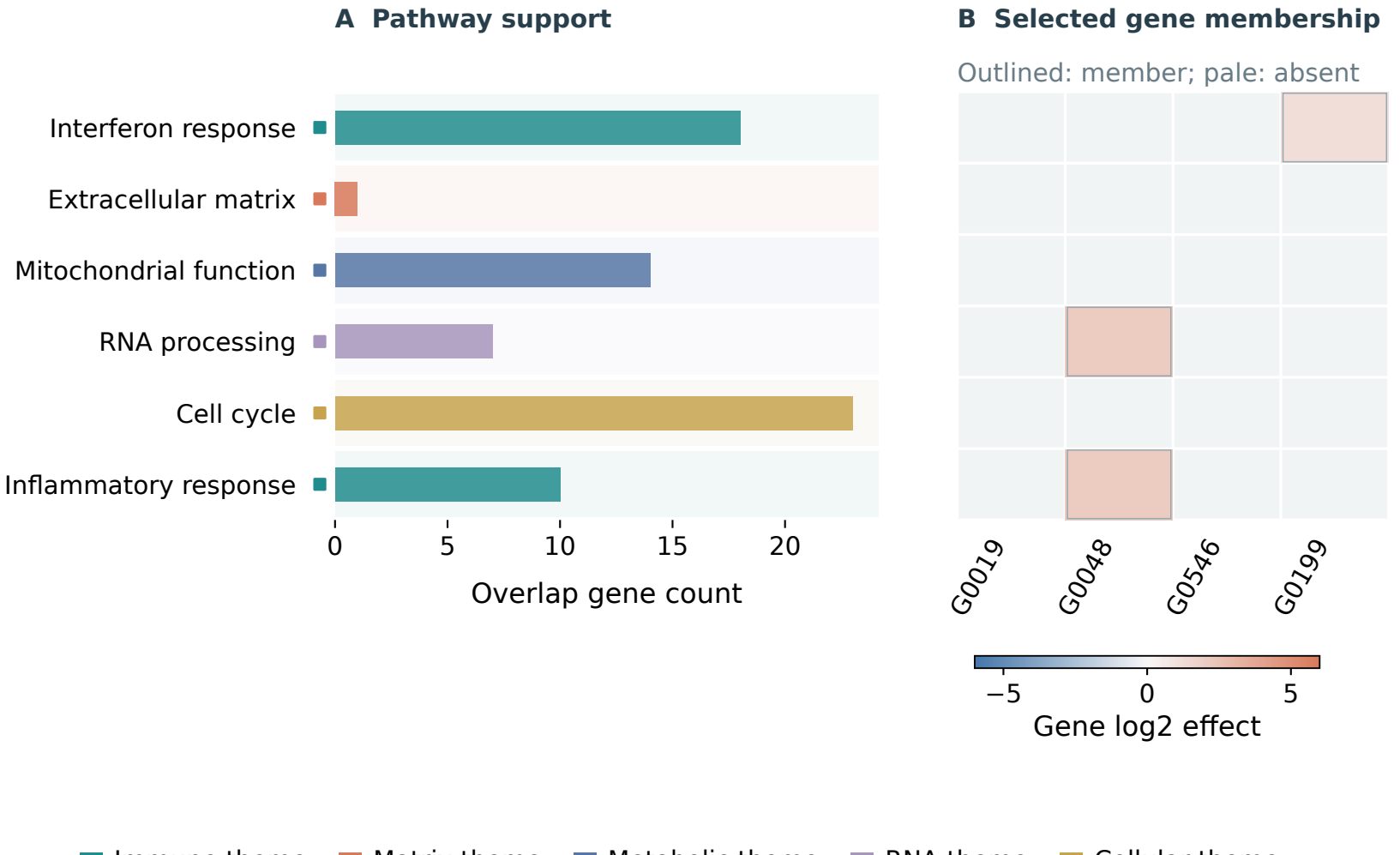
Myeloid · GSEA pathway responses and gene support

Gene effects: Treated versus Control / ranked universe: 1200



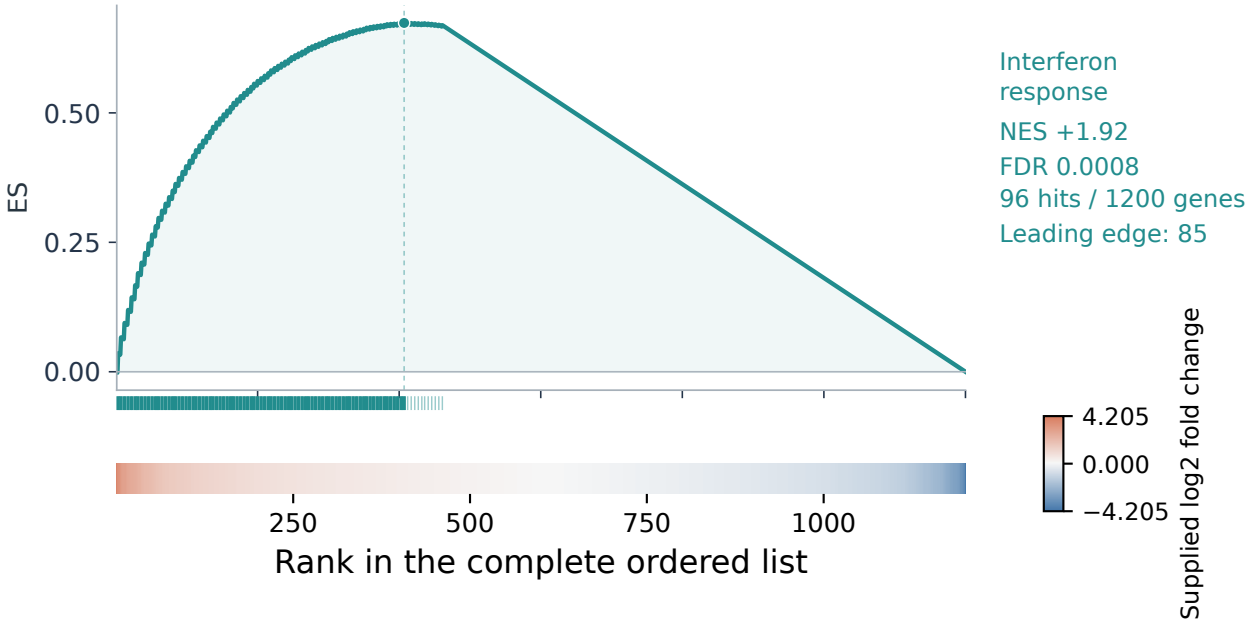
Myeloid · ORA membership and selected gene effects

Gene effects: Treated versus Control / query: 148 / tested: 1200



Interferon response · complete GSEA evidence

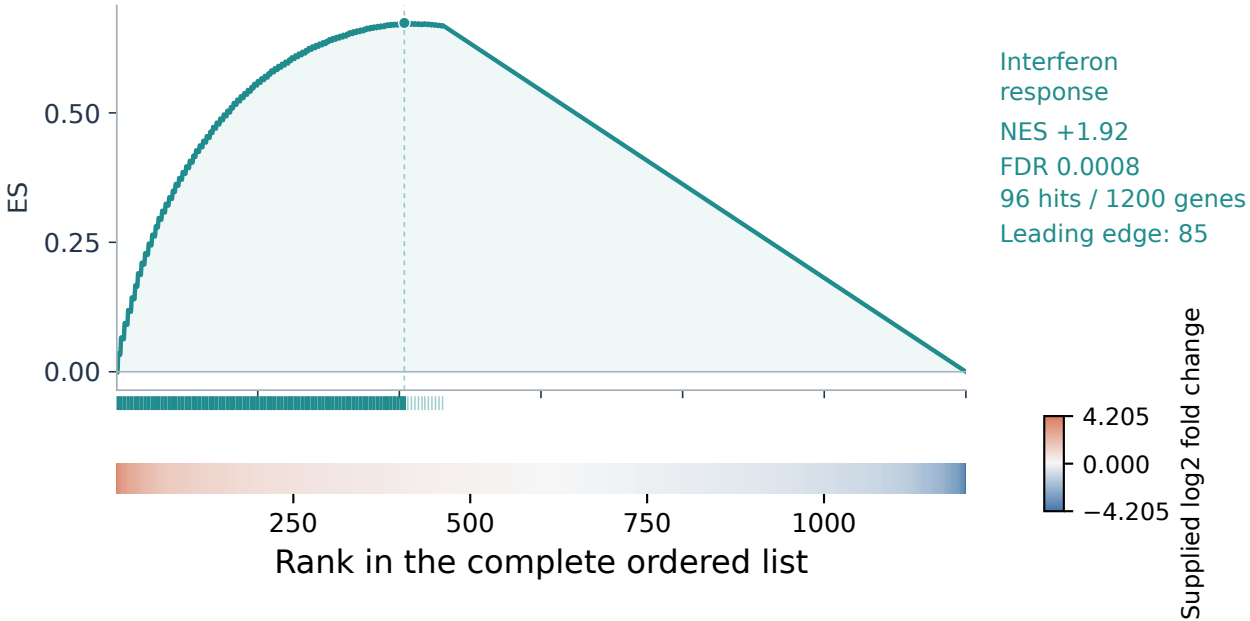
Treated versus Control / illustrative / exact supplied enrichment tracks



Hit rugs retain exact ranks; bold hits are supplied leading edge. Fictional complete gene set v0.9. No ES, NES, FDR or membership recomputation.

Myeloid · Interferon response · complete GSEA evidence

Treated versus Control / illustrative / exact supplied enrichment tracks



Hit rugs retain exact ranks; bold hits are supplied leading edge. Fictional complete gene set v0.9. No ES, NES, FDR or membership recomputation.

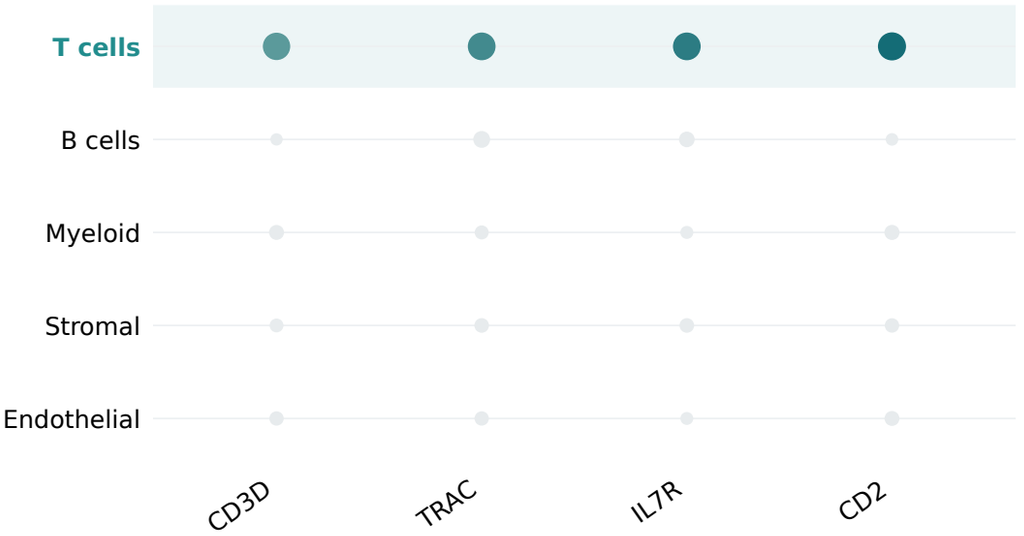
T cells · requested two-column marker block

T cells / 226 of 900 displayed cells / 4 selected markers

A Population location



B Marker specificity (mean)



C CD3D



D TRAC

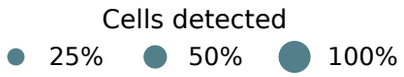


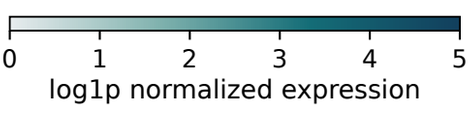
E IL7R



F CD2

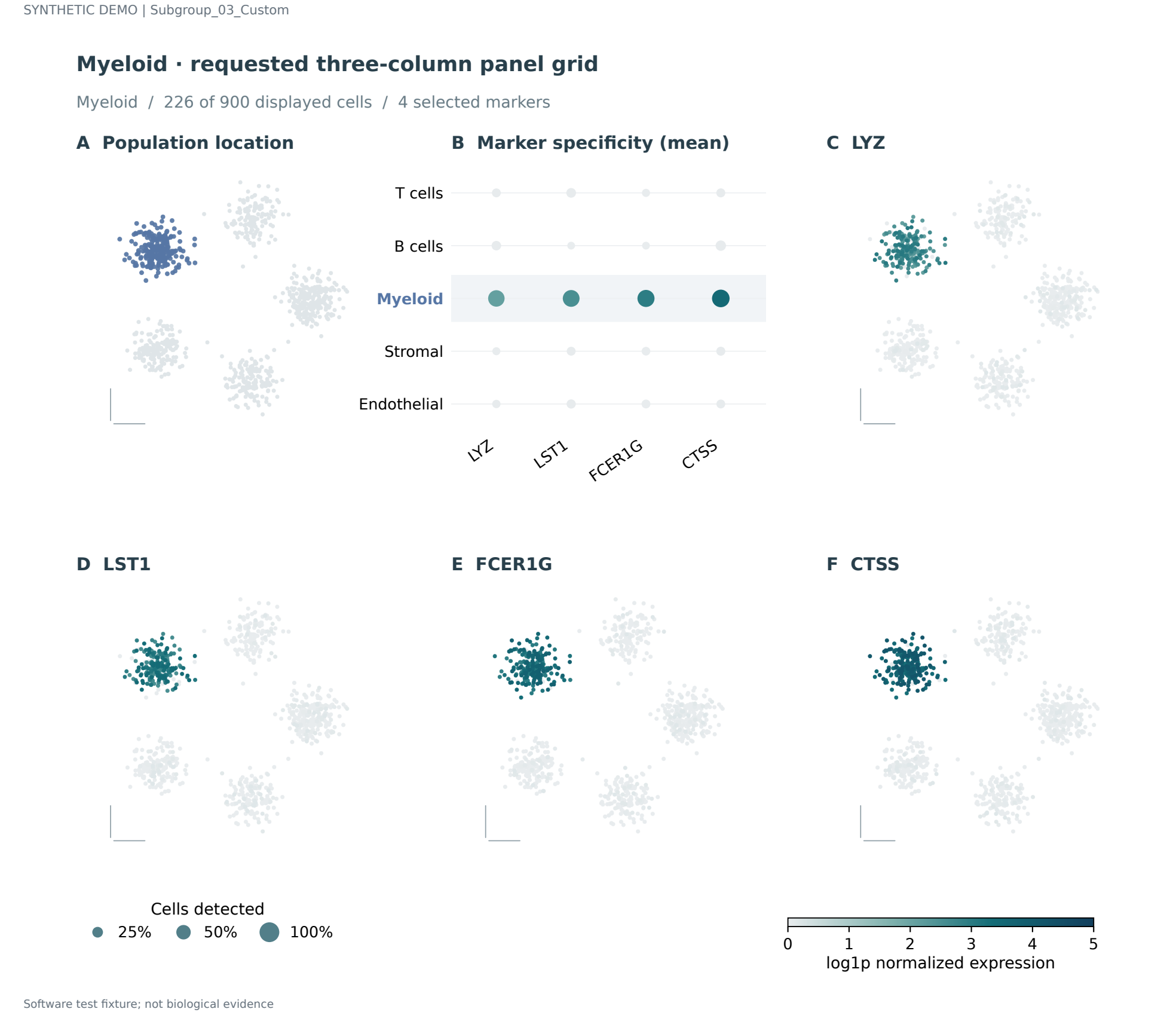






Myeloid · requested three-column panel grid

Myeloid / 226 of 900 displayed cells / 4 selected markers



GSEA · requested horizontal pathway comparison
Treated versus Control / illustrative · Fictional complete gene set v0.9
Bold hits = supplied leading edge · rank metric: Supplied log2 fold change

